Diversity-Connected Graph Convolutional Network for Hyperspectral Image Classification

Yun Ding, Yanwen Chong, Shaoming Pan, and Chunhou Zheng

***Abstract*— Hyperspectral image (HSI) classification methods based on the graph convolutional network (GCN) have received more attention because they can handle irregular regions by graph encoding techniques. However, GCN-based HSI classification methods are highly sensitive to the quality of the graph structure. Its performance degrades in the case of underdeveloped graphs because it cannot excavate the intrinsic adjacency relationships. Thus, it is necessary to improve the quality of graph structure in GCN-based methods. In this article, a novel diversity-connected GCN (DCGCN) method is proposed to improve the quality of the graph structure for HSI classification, and its basic idea can be adopted by other GCN-based methods. First, the potential neighbors are excavated by performing topological extensions based on the given graph. The diversity of surrounding neighbors is maintained by adaptively smoothing operation via a global threshold value from Kullback–Leibler (KL) divergence to eliminate weak interclass connections caused by weakly spectral variability. Second, another key connectivity restriction is imposed on the diverse neighbors to further refine the ambiguous connections of hard samples aiming at removing strong interclass connections where the spectral information is heavily confounded. Finally, the DCGCN method is analyzed theoretically to demonstrate its low-pass filter property. The comprehensive experiments demonstrate the effectiveness of the proposed DCGCN method and the basic idea of the diversity-connected graph in terms of overall accuracy (OA), kappa coefficient (KC), and average accuracy (AA) indices.**

***Index Terms*— Diverse neighbors, graph connectivity, graph convolutional network (GCN), hyperspectral image (HSI) classification.**

NOMENCLATURE

v_i	Pixel node.
N_{v_i}	Surrounding neighbors of the node v_i.
$\bar{\mathbf{H}}^{(k)}$	Feature information from the kth layer.
$\mathbf{D}$	Degree formation of nodes.
$\mathbf{L}$	Symmetric positive semidefinite Laplacian matrix.
$\bar{\mathbf{A}}$	Adjacency similarity graph with self-loops.
$\mathbf{A}$	Adjacency similarity graph.
$\mathbf{X}$	Initial feature information.
$\mathbf{\Theta}^{(k)}$	Weight parameter from the kth layer.
$\mathbf{Z}$	Prediction result.
$\mathbf{Y}$	Label indicator.
$\mathbf{G}$	Initial graph structure.
$\mathbf{W}$	Topological extension graph.
$\bar{\mathbf{W}}$	Adoptive smoothing matrix.
$\hat{\mathbf{W}}$	Final graph.
C	Probability density function (pdf) of the core node.
S	PDF of surrounding neighbors.
Ω	Normalized graph feature space.
λ_a	Graph-specific parameter.
μ	Length of the diverse neighbors.

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I. INTRODUCTION

HYPERSPECTRAL sensors sample reflected solar radiation from the Earth's surface in a portion of the spectrum extending from the visible region to the near-infrared and mid-infrared (wavelengths between 0.3 and 2.5 um). The spectrum contains hundreds of narrow (around 10 nm) contiguous bands [1]. This high spectral resolution can offer a unique ability to distinguish subtle spectral features of different land covers. Due to its unique spectral advantage, hyperspectral image (HSI) has been successfully applied in many fields, such as target detection [2], spectral unmixing [3], and classification [4]. Among these applications, HSI classification has received much attention because of its advantages in precise agriculture, urban investigation, and environment monitoring [5], [6], [7].

Different from general land-use classification methods that aim to understand the major contents of the scene [8], HSI classification aims to assign a semantic class label to each pixel [9], which is a hot topic in the HSI community. In the early stage of HSI classification research, various traditional methods based on hand crafting have been proposed to explore pixelated classification and considerable classification results have been obtained. For example, methods based on principal component analysis (PCA) [10] and independent component analysis (ICA) [11] were proposed to extract effective spectral features. Their purpose was to transform the original feature space of input data into an independent feature space to distinguish different semantic classes. However, spectral information cannot provide feature information for the separation of different semantic classes due to the large spatial variability of spectral signatures caused by many factors such as illumination, environment, and atmospheric. Therefore, several methods based on spatial-spectra, including sparse representation (SR) [9], low-rank representation (LRR) [12], and Markov random field (MRF) [13], were proposed

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to alleviate this phenomenon by exploring rich spatial information.

Compared to the above traditional methods, deep learning methods combine feature extraction with the downstream tasks to form a unified framework, automatically learning more suitable features for coping with various situations [4], [14]. Generally, methods based on convolutional neural networks (CNNs) are representative classification methods for deep learning models. The existing CNN models in HSI classification can be divided into two categories: spectral CNNs and spatial-spectral CNNs. Spectral CNNs usually refer to 1-D CNNs, which aim to refine the effective spectral features for each pixel along with the direction of spectral bands. The input of the system is a pixel vector of hyperspectral data, and the output of the system is the label of the corresponding pixel vector. Moreover, it mainly consists of several simple convolutional and pooling layers [15], limiting their high-level feature representation ability because it lacks spatial information in 1-D CNN models. To alleviate this problem, the 2-D CNN models were designed to explore effective spatial contextual information by selecting square windows [16], [17]. Different from 1-D CNNs and 2-D CNNs, 3-D CNNs were proposed to extract spectral and spatial features simultaneously from the original HSI [18], [19]. Unfortunately, the tuning of 3-D convolutional kernels often costs much time or needs a large number of parameters [20]. In order to alleviate this problem, many studies have proposed designing two-branch networks to extract spectral and spatial features. In [21] and [22], a parallel two-branch framework was proposed to extract spectral features by using 1-D CNN and spatial features using 2-D CNN, which has a significantly lower computational burden than 3-D CNNs.

Most CNN-based methods with a fixed kernel shape cannot effectively preserve the details of irregular cross-class edges and small irregular regions in land covers [23]. In order to address this problem, methods based on graph convolutional networks (GCNs) were applied to complex HSI because their flexible graph structure can better model complex irregular regions [24]. Based on the superiority of GCN models, the spectral-spatial GCN (S^2GCN) [25] method was proposed to combine intrinsic spectral information and local spatial window information for HSI classification. However, the graph structure of the S^2GCN method is essentially a local graph. It cannot capture the comprehensive contextual information of HSI, degrading the global inference performance. Thus, global GCN-based models such as global consistent GCN (GCGCN) [26] and non-local GCN [27] were presented to capture the underlying rich contextual information by exploring global high-order consistency. Despite the good classification performance of the above GCN models, their computational burden in network architecture exponentially increases with the number of GCN layers due to the neighborhood expansion in the pixel-level graph structure. Some studies proposed multiscale dynamic GCN (MDGCN) [24] and context-aware dynamic GCN (CAD-GCN) [28] to build the graph structure based on superpixels to achieve superpixel-level neighbor information aggregation based on the GCN model instead of pixel-level neighbor aggregation. Since pixel-level data representation and superpixel-level data representation are incompatible in a single network, the CNN-enhanced GCN (CEGCN) [29] method was proposed to enable CNN with pixel-level representation and GCN with superpixel-level representation to collaborate in a single network. The above GCN-based methods are transductive learning and cannot handle out-of-sample data. Based on this problem, minibatch GCN (MiniGCN) and its improved end-to-end fusion network (FuNet) [30] between the miniGCN and CNN were proposed to achieve HSI classification.

However, in various complex HSI environments, the above GCN-based HSI classification methods cannot guarantee the quality of the graph structure. In fact, the quality of the graph structure in GCN-based methods directly determines the quality of the feature propagation in the cascade network, which is critical for downstream tasks [31]. For those GCN-based HSI classification methods, it is a challenge to guarantee the quality of the graph structure due to the characteristic of HSI and methods of the constructed graph. The main reasons are as follows. First, most real HSIs usually contain instrumental noise [32], which leads to the generation of a noisy graph structure. The noisy graph propagates noise to surrounding neighbors, affecting the discrimination of the embedding feature space. Second, the variability of spectral signature in intraclass pixels leads to the degradation of the graph quality. The spectral variability is always existing in HSI, which leads to the unconnected graph structure of intraclass pixels. Third, the interclass similarity of spectral information leads to the redundant graph structure because of the low spatial resolution and the instinct similarities between different land covers. Specifically, the graph structure usually contains the ambiguous connections of the hard samples located at the region of severe spectral confusion when constructing the intraclass graph structure. Fourth, the graph structure is usually obtained by hand-designed methods or learning paradigms using complex feature information, which is difficult to generate a discriminant graph [14] and limits the corresponding inference performance of GCN-based methods in HSI classification. In summary, GCN-based methods are highly sensitive to the quality of the graph structure and usually require a perfect graph structure to learn informative representations [33]. On this basis, it is a challenge to improve the quality of the given graph structure in GCN-based methods.

For further demonstrating the motivation, the related experiments with initial graphs in GCN-based HSI classification methods are conducted to prove the rationality of the motivation. Specifically, the spectral clustering results [34] are used to evaluate the quality of graph structure from GCN-based methods because the spectral clustering performance depends on the quality of the constructed graph to a certain degree. In other words, the clustering results will be good when intraclass connections of the initial graph between HSI pixels are guaranteed, and vice versa. Consequently, the spectral clustering results can prove the quality of the graph structure to some degree, and the corresponding analysis can be observed in Fig. 1.

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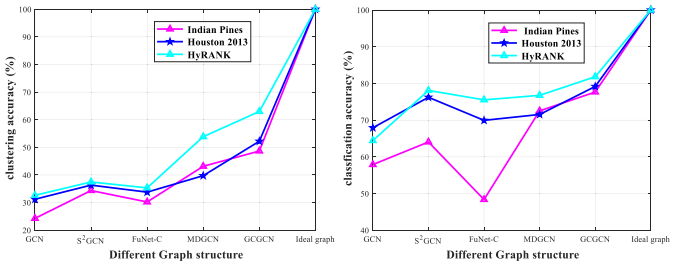


Fig. 1. Clustering and classification results with different original graph structures on three HSI datasets. From (a), the clustering accuracy on different graph structures is around 50%. However, the clustering result of ideal graph are 100%. The ideal graph refers to the graph constructed by the hand-crafting design method that preserves intraclass connections and fully removes interclass connections. Consequently, the clustering results on three datasets demonstrate that the quality of the graph structure is usually poor due to the complex spatial-spectral characteristics of HSIs. From (b), the trend of the classification performance with different graph structures is generally consistent with that of the clustering results. Consequently, the classification results on three datasets further demonstrates that the importance of the graph structure on GCN-based methods. Based on above analysis, it is necessary to improve the quality of graph structure in GCN-based methods from the perspective of topological view inspired by ideal graph.

Based on the above analysis, existing GCN-based methods usually obtain the noisy graph structure that refers to interclass connections in the graph structure. However, existing solutions always focus on the construction of the network architecture to alleviate the influence of the noisy graph structure. In fact, the noisy graph based on GCN-based methods often aggregates the neighbor information containing the noise neighborhood structure via a massage passing mechanism due to complex characteristics of HSIs, which affects the final reasoning performance. Therefore, it is necessary to consider a method independent of the network architecture to process the initial poor graph so as to hinder the propagation of noise information. In addition, most GCN-based methods only consider neighborhood relationships in terms of spectral-spatial feature information. The mining of the effective neighbors from the topological view is neglected, making it difficult to mitigate the influence of spectral variability in graph construction. The underlying effective neighbors from a topological view should be explored to remove the influence of severe spectral variability. Consequently, we proposed the key connectivity which is connectivity strength to remove the influence of severe spectral variability in graph structure. The graphical explanation of the topological neighbor restriction is shown in Fig. 2. The basic idea is that the stronger the connectivity strength μ is, the greater the probability of belonging to the same class is. The class information of pixels can be ensured by the strength of the parameter μ in key connectivity restriction to a certain degree. When the distributions of surrounding neighbors are different and the labels of surrounding neighbors are consistent, the GCN will aggregate beneficial information from surrounding neighbors of intraclass pixels for downstream work. Finally, the theoretical analysis in the graph construction is also lacking for GCN-based HSI classification method.

In this study, a novel diversity-connected GCN (DCGCN) for effective HSI classification is proposed. It mainly contains two steps. First, the coarse neighborhood selection is performed using an adaptive smoothing operation inspired by the Kullback–Leibler (KL) divergence theory to ensure the diversity of the surrounding neighbors for obtaining the effective information. Then, the fine-grained neighbor selection is performed using the key connectivity of diverse neighbors to further maintain the quality of diverse neighbors and obtain the final graph structure. Notably, the DCGCN method is a post-processing framework independent of the network architecture, which improves the quality of the given graph structure by fully utilizing the topological information of the graph structure. The contributions of this study are summarized as follows.

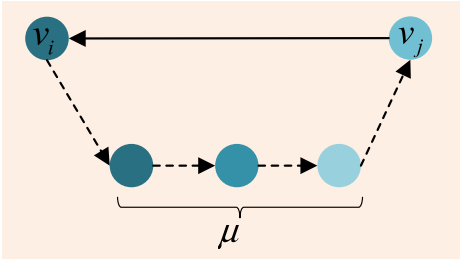


Fig. 2. Shades of color denotes the different feature diversity of the same class. The solid line denotes the direct neighbor, and the dotted line denotes the underlying μ neighbors, called key connectivity from a topological view. Nodes v_i and v_j are truly beneficial neighbors when there is the underlying μ neighbors between them.

1) A novel plug-and-play post-processing framework independent of the network architecture DCGCN is proposed, which improves the quality of the given graph structure by fully utilizing topological information. Moreover, the performance of the DCGCN can be guaranteed by the KL divergence and the low-pass filter theory.
2) Different from the other GCN-based methods that focus on building the effective network, the proposed method aims to formulate a reasonable rule that considers both the diversity and the connectivity of the graph structure. In this way, the quality of existing GCN-based methods can be enhanced for effective HSI classification.
3) The corresponding theoretical analysis of the DCGCN method is performed to improve the quality of the graph structure. Moreover, the comprehensive experiments on three real HSI datasets demonstrate the effectiveness of the basic idea underlying the existing GCN-based methods of plug-and-play.

II. Related Work

The basic idea of the proposed DCGCN model is closely related to previous GCN-based models. In this section, some representative GCN-based HSI classification methods and the principles of GCN models are reviewed.

A. GCN for HSI Classification

Different from CNN-based methods that operate on regular Euclidean data to extract the local stationarity property of the input data such as images (2-D grid) and text (1-D sequence) [35], GCNs can encode irregular or non-Euclidean data based on spectral graph theory [36]. Bruna et al. [37] first

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defined spectral graph convolution on the Fourier domain by computing the eigen decomposition of graph Laplacian, but the computation is large. Defferrard et al. [36] proposed to approximate the filters through Chebyshev expansion of the graph Laplacian, avoiding the computation of the Laplacian eigenvectors. Subsequently, Kipf and Welling [38] proposed a GCN model that simplified the previous method by restricting the filters to operate in a 1-step neighborhood around each node. Based on the success of GCN-based models on irregular and regular data [39], [40], Qin et al. [25] first proposed the S^2GCN method for semi-supervised HSI classification. In order to address the local limitation, Ding et al. [26] proposed a globally consistent GCN method, and Mou et al. [27] proposed a non-local GCN method for HSI classification. Hong et al. [30] introduced the FuNet model for HSI classification. The above classification methods belong to pixel-level GCN-based methods. Some researchers proposed superpixel-level GCN-based methods to explore rich spatial-contextual information aggregation in the massage passing mechanism. For example, Wan et al. [24], [28], and [41] proposed the MDGCN, CADGCN, and multilevel graph learning network (MGLN) methods for HSI classification, respectively. Aiming to explore the effective collaboration of the CNN and GCN models, Liu et al. [29] and [42] proposed the CEGCN model as well as the multilevel superpixel structured graph U-Net (MSSGU) model, respectively. In addition, the multiple feature fusion model termed attention multihop graph and multiscale convolutional FuNet (AMGCFN) [43] is proposed to extract the multilevel information of HSI. The adaptive filters and aggregator fusion (AF2GNN) [44] method was developed for HSI classification. The semantic-aware graph network (SAGN) [45] was proposed to explore various hidden semantics in high-resolution remote sensing image classification. Moreover, some researchers utilize the hypergraph neural network to describe complex spectral characteristics. For example, the structure-preserved hyper GCN (SPHGCN) method [46] was proposed to reveal the essential resemblance of multiple pixels in nonlinear spectral feature space. The feature fusion hypergraph neural network (F^2HNN) [47] was proposed to represent multimodal features in HSI.

However, these studies pay less attention to the quality of the graph structure in GCN-based methods relying on the network architecture. Moreover, these studies failed to provide a unified and detailed framework from the perspective of the topological neighbors, making them incapable of obtaining valuable information from surrounding neighbors to improve the quality of the graph. These studies also ignored insights into the graph construction in terms of theoretical framework. Based on the previous studies, the DCGCN model proposed in this study is independent of the network architecture, allowing the improvement of the graph quality in GCN-based models.

B. GCN Principle

The basic GCN model consists of three steps: feature aggregation, feature transformation, and classification. Each of these steps is described below.

1) Feature Aggregation: The basic idea of feature aggregation is to obtain the information of surrounding neighbors for self-enrichment so as to achieve intraclass feature smoothing [48]. At the beginning of each layer, the new features $\mathbf{h}_i$ of each node v_i at the kth layer can be calculated as the weighted average of its local neighbors

$$\bar{\mathbf{h}}_i^{(k)} \leftarrow \frac{1}{d_i+1}\mathbf{h}_i^{(k-1)} + \sum_{j\in \mathrm{N}_{v_i}} \frac{a_{ij}}{\sqrt{(d_i+1)(d_j+1)}}\mathbf{h}_j^{(k-1)} \quad (1)$$

where $\mathbf{D} = \mathrm{diag}(d_1, \ldots, d_N)$ denotes the degree matrix, $d_i = \sum_{j=1}^{N} a_{ij}$ denotes the row-sum of the adjacency matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$; $a_{ij} \in \mathbf{A}$ represents the edge weight between nodes v_i and v_j; N denotes the number of nodes in this graph; N_{v_i} denotes the surrounding neighbors of the node v_i. For clarity, (1) can be expressed by the matrix operation

$$\bar{\mathbf{H}}^{(k)} \leftarrow \bar{\mathbf{D}}^{-1/2}\bar{\mathbf{A}}\bar{\mathbf{D}}^{-1/2}\mathbf{H}^{(k-1)} \quad (2)$$

where $\bar{\mathbf{A}} = \mathbf{A} + \mathbf{I}$, $\mathbf{I}$ is the identity matrix; $\bar{\mathbf{D}}$ represents the degree matrix of the adjacency matrix $\bar{\mathbf{A}}$; $\mathbf{H}^{(0)}$ denotes the original feature matrix $\mathbf{X}$. Equation (2) is the graph convolution, which smoothens the hidden representations by aggregating neighboring information and encouraging similar predictions among locally connected nodes.

2) Feature Transformation: After local smoothing of all nodes using feature aggregation via (2), the learned weight matrix $\mathbf{\Theta}^{(k)}$ of a kth hidden layer is applied to new features $\bar{\mathbf{H}}^{(k)}$ to obtain embedding features. Afterward, the weighted features are passed through a nonlinear activation function such as ReLU to output the final feature representation as

$$\mathbf{H}^{(k)} \leftarrow \mathrm{ReLU}(\bar{\mathbf{H}}^{(k)}\mathbf{\Theta}^{(k)}) \quad (3)$$

where $\mathrm{ReLU}(\cdot) = \max(0, \cdot)$; $\mathbf{\Theta}^{(k)}$ is the learned weight matrix.

3) Classification: The final layer of GCN aims at predicting the labels of all nodes using the softmax classifier. Let $\mathbf{Z} \in \mathbb{R}^{N\times C}$ be the class prediction results of N nodes, where $\mathbf{Z} = \mathrm{soft\,max}(\mathbf{H}^{(k)})$ and C denotes the number of classes. For the semi-supervised node classification task, the following cross-entropy error function is used to evaluate the performance of the obtained GCN model on all labeled nodes:

$$L_{\mathrm{GCN}} = -\sum_{i\in\Upsilon}\sum_{j=1}^{C}\mathbf{Y}_{ij}\ln\mathbf{Z}_{ij} \quad (4)$$

where Υ denotes the set of node indices with labels and $\mathbf{Y}$ denotes the corresponding label indicator. In addition, there are many symbols, which may confuse readers. For the convenience of reading, these symbols are listed in the Nomenclature.

III. Proposed Method

A. Framework of the Proposed Method

The flowchart of the proposed model is shown in Fig. 3. It mainly consists of two critical constraints: the diversity of the surrounding neighbors and the key connectivity of the diverse neighbors. First, given a pixel- or superpixel-based graph structure $\mathbf{G}$, a topological graph $\mathbf{W}$ can be obtained to excavate the diversity of the surrounding neighbors. The diversity quality of the surrounding neighbors is further preserved using the adoptive smoothing matrix $\bar{\mathbf{W}}$ to remove weak

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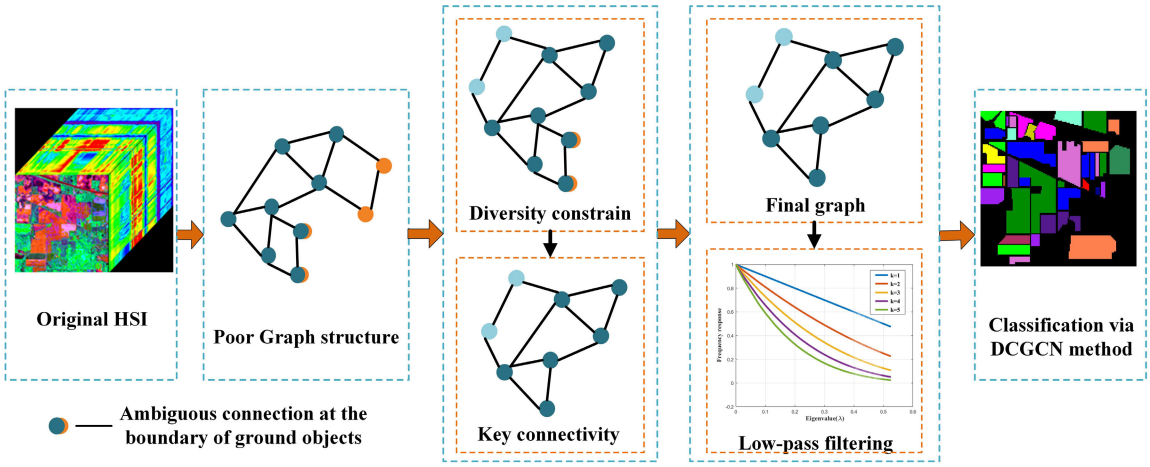


Fig. 3. Flowchart of the proposed DCGCN method to achieve information aggregation from surrounding neighbors. Different colors of different circular nodes are from different classes. Moreover, the hue of the color denotes the diversity of the surrounding neighbors.

interclass connections of graph structure via a global threshold value λ_a, which is from the analysis of Theorem 1. The diversity quality refers to the positive intraclass information from surrounding neighbors, and the negative diversity refers more to interclass information of surrounding neighbors. Afterward, a key connectivity constraint imposed on the adaptive smoothing matrix $\bar{\mathbf{W}}$ to obtain the final graph $\hat{\mathbf{W}}$. It guarantees the key connectivity of the diverse neighbors and remove strong interclass connections of graph structure correcting the connectivity of the hard samples caused by severe spectral confusions. Overall, the adaptive smoothing matrix $\bar{\mathbf{W}}$ aims at perform the selection of the coarse neighbors, and explore the effective surrounding neighbors from the perspective of the topology, and remove the common weak interclass connections in adjacent neighbors. The ambiguous connections of hard samples referring to strong interclass connections caused by severe spectral confusions can be corrected by the key connectivity. The proposed two restrictions aim to preserve positive intraclass connections and remove negative interclass connections in improving the quality of the given graph structure.

B. Selection of Diversity Neighbors

Given a graph $\mathbf{G}$ without guaranteed quality, a topologically consistent extension is first performed to reconstruct the new graph instead of relying on spectral-spatial features of HSI. The reason for using the topological extension is that the topological extension of the adjacencies can excavate more potential neighbors to learn reliably informative representations when the original feature information is corrupted by noise. The topological extension of the given graph $\mathbf{G}$ can be expressed as

$$\min_{w} \sum_{r}^{N} \left(\frac{1}{2} \sum_{i,j=1}^{N} g_{ij} \|w_{ri} - w_{rj}\|^2 + \eta \sum_{i=1}^{N} \|w_{ri} - I_{ri}\|^2 \right) \tag{5}$$

where $w_{ij} \in \mathbf{W}$ is the obtained topological graph based on the given graph $g_{ij} \in \mathbf{G}$; $I_{ri} \in \mathbf{I}$ denotes the element of the rth row and the ith column in the identity matrix $\mathbf{I}$; and η denotes the balanced parameter, and the parameter η is not too large so as to avoid approximating the identity matrix. The basic idea of (5) is that for any two nodes, their spectral-spatial similarity may be low, their topological relevance will be high. If they are connected by consecutive neighbors in a manifold structure, their topological relevance to each other becomes high. Thus, if nodes i and j have high similarity, their topological relevance to any other nodes is assumed to be consistent [49]. Because (5) is independent of different r, the following equation is proposed to handle each r:

$$\min_{w_r} \sum_{r}^{N} \left(\frac{1}{2} \sum_{i,j=1}^{N} g_{ij} \|w_{ri} - w_{rj}\|^2 + \eta \sum_{i=1}^{N} \|w_{ri} - I_{ri}\|^2 \right) \tag{6}$$

where $\boldsymbol{w}_r$ is the rth column of matrix $\mathbf{W}$. Afterward, the derivative of (6) with respect to r can be obtained

$$\mathbf{L}\boldsymbol{w}_r + \eta(\boldsymbol{w}_r - \mathbf{I}_r) = 0 \tag{7}$$

where $\mathbf{L} \in \mathbb{R}^{N \times N} = \mathbf{D} - \mathbf{G}$ is the Laplacian matrix of matrix $\mathbf{G}$ and $\mathbf{I}_r$ is the rth column of $\mathbf{I}$. Then, the optimal representation is obtained

$$\mathbf{W} = \left(\mathbf{I} + \mathbf{L}/\eta\right)^{-1}. \tag{8}$$

For a given poor graph $\mathbf{G}$, its topological extension $\mathbf{W}$ can be obtained through (8). Although $\mathbf{W}$ can excavate more potential neighbors to enrich the diversity of adjacent neighbors, it may maintain a correlation with interclass neighbors because of the complex spectral-spatial characteristics. Next, a coarse selection of the surrounding neighbors is conducted to remove interclass connections and preserve the diversity of neighboring information. When the effective features of surrounding neighbors are different, the GCN model can obtain a powerful representation because it can capture the different distribution of neighbors [50]. The effective feature refers to the intraclass features with diverse distributions of surrounding neighbors, which provide positive information for downstream task. In order to quantify the diversity of surrounding neighbors, spectral features, or spatial information in HSI could be a choice. However, due to the great variability of spectral features within classes and the limitations of the

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local properties of spatial information, the diversity of the surrounding neighbors should be explored by utilizing topological relationships instead of only using spectral or spatial information. Based on the above analysis, the effective diversity of surrounding neighbors is explored from the perspective of the topological property of the graph structure. To quantify the diversity of the surrounding neighbors, the KL [51] divergence is adopted to quantify the topological relationship of surrounding neighbors, which can be expressed as

$$D_{\mathrm{KL}}(S\|C) = \int_{x\in\Omega} S(x)\cdot \log \frac{S(x)}{C(x)}dx \tag{9}$$

where C denotes the pdf of the node itself v_i; S denotes the pdf of surrounding neighbors; Ω denotes the normalized graph feature space, i.e., $\Omega = [0,1]^d$; and d denotes the number of data points. The KL divergence is a measure of information loss when the graph structure distribution of surrounding neighbors is used to approximate the graph structure distribution of the node itself. The basic idea of (9) is used to evaluate the consistency between the distribution of the surrounding neighbors and the distribution of the node in the graph structure space. In other words, it can be used to evaluate the effective information gained from surrounding neighbors in the node. If the distribution of the node itself is equal to that of the surrounding neighbors in graph structure space, there is no information gain, that is $D_{\mathrm{KL}}(S\|C) \to 0$. As a result, the surrounding neighbors cannot enrich the representation of the node itself. Furthermore, the node itself can learn more from surrounding neighbors when the distributions are different in graph structure space. The graph representation of the surrounding neighbors is diverse, and extra effective information gain can be obtained by the node itself. In practice, the real distributions of the core node and the surrounding neighbors are unknown. They can be estimated by nonparametric methods with a set of samples. To further preserve the divergence of the topological neighbors, Theorem 1 is proposed based on KL divergence.

Theorem 1: For a given graph $\mathbf{G}$ with the set of graph representation features in space, such as, $\Omega = [0,1]^d$, the effective surrounding information gain $D_{\mathrm{KL}}(S\|C)$ is positively correlated with its similarity $1/a$, such as $D_{\mathrm{KL}}(S\|C) \sim 1/a$. In particular, $D_{\mathrm{KL}}(S\|C) \to \infty$ when $1/a \to \infty$ and $a \to 0$.

The proof can be found in Appendix. According to Theorem 1, a larger $D_{\mathrm{KL}}(S\|C)$ indicates a smaller neighbor similarity a. Thus, to preserve the effective diversity of surrounding neighbors of each core node in graph structure space, an adaptive smoothing $\bar{\mathbf{W}}$ matrix based on the topological graph $\mathbf{W}$ is used to maintain the quality of the diversity of each core node. The adaptive smoothing matrix can be defined as

$$\max\left\{\|\mathbf{W}_i - \bar{\mathbf{W}}_i\|_1 < \lambda_a\right\}. \tag{10}$$

The basic idea of (10) aims to preserve the diversity quality of the surrounding neighbors by removing weak interclass connections of the nodes in graph structure through l_1 norm, where $\bar{\mathbf{W}}_i$ denotes the adaptive smoothing matrix to preserve the neighbor diversity of the ith node; $\|\cdot\|_1$ denotes the one-norm; λ_a represents a graph-specific parameter that is a suitable constant with $\lambda_a > 0$ to preserve the diversity quality of every core node i. Specifically, (10) preserves the intraclass similarity of $\bar{\mathbf{W}}_i$ by keeping some of the larger similarity values using the graph-specific constant λ_a via l_1-norm because the maximum difference between the adaptive smoothing matrix $\bar{\mathbf{W}}_i$ and the topological extension matrix $\mathbf{W}_i$ is no more than the intraclass similarity measurement λ_a. Consequently, the larger similarity in the topological extension matrix $\mathbf{W}_i$ can be preserved. Next, (10) removes the smaller similarity value of topological extension matrix $\mathbf{W}_i$ by using the graph-specific threshold λ_a with the l_1-norm because smaller similarity values cannot satisfy the requirement of potential maximum differences between graph structure. So, smaller weight values with weak interclass connections of graph structure can be effectively removed, and larger weight values that are more likely belonging to intraclass connections are preserved. In other words, the diversity quality of the surrounding neighbors in the adaptive smoothing matrix $\bar{\mathbf{W}}$ can be guaranteed by using a suitable parameter λ_a. Indeed, how to balance the quality of graph structure and the diversity of the surrounding neighbors is an important issue. Overall, the robustness of the graph-specific parameter λ_a needs to be considered because the spectral variation and the noise in real HSI data are common. Moreover, the graph structure of these available training samples is explicit, and the compactness within every class is relatively close. Based on the prior knowledge of the existing training samples, the global statistical properties of intraclass graph representation can be used to a faithful threshold to preserve the diversity quality of every core node. The suitable graph-specific parameter λ_a can be defined as

$$\lambda_a = \arg\min_{l_i} \frac{\sum_i \|\mathbf{W}^{l_i}\|_1}{|\varepsilon|} \tag{11}$$

where l_i denotes the ith class for training; $\mathbf{W}^{l_i} \in \mathbb{R}^{|l_i|\times|l_i|}$ denotes the subgraph matrix for training samples belonging to the ith class; $|\varepsilon|$ and $|l_i|$ denote the number of non-zero edge in matrix $\mathbf{W}^{l_i}$ and the number of training samples in the ith class, respectively. In summary, for a given poor graph structure $\mathbf{G}$, a diverse neighbor structure $\mathbf{W}$ is first obtained from the perspective of the topological extension, and a new neighbor similarity matrix $\bar{\mathbf{W}}$ is formulated by (10), which preserves the quality of the diverse neighbors by removing common noise in a robust way.

C. Connectivity of Diverse Neighbors

According to Theorem 1, the adaptive smoothing matrix $\bar{\mathbf{W}}$ is obtained through (10). With the above steps, the diversity of the surrounding neighbors allows the core node to learn more quality information. Although the influence of the spectral variation in real HSI are mitigated to a certain degree, the ambiguous connections of the hard samples, such as positioning in severe spectral confusions, cannot be carefully corrected due to the involvement of the complex interclass information at low spatial resolution. Those GCN-based methods ignore the above problems and cannot further refine the graph structure to mine its intrinsic property from a topological view. Thus, the key connectivity of diverse neighbors is explored to ensure the

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quality of the hard sample connectivity and eliminate strong interclass connections of surrounding neighbors. In terms of the node v_i, the key connectivity of diverse neighbors can be defined as follows:

$$\Phi\left(v_i, \mathbb{N}_{\lambda_a}(v_{k_1})\right) \times \cdots \times \left(\prod_{m=2}^{\mu} \Phi\left(v_{k_{m-1}}, \mathbb{N}_{\lambda_a}(v_{k_m})\right)\right) \times \Phi\left(v_{k_\mu}, \mathbb{N}_{\lambda_a}(v_j)\right) \times \Phi\left(v_j, \mathbb{N}_{\lambda_a}(v_i)\right) = 1 \quad (12)$$

where Φ denotes the indicator function; $\Phi(v_i, \mathbb{N}_{\lambda_a}(v_j)) = 1$ denotes that the node v_i satisfy the requirement of the diversity neighbor for the node v_j; and μ denotes the number of underlying diverse neighbors between nodes v_i and v_j. In addition, μ also represents the length of the diverse neighbors, which allows the effective diverse information that the node v_i benefits from the neighboring node v_j. The intuitive explanation of key connectivity is to preserve consecutive neighbors in an underlying structure. For convenience, the basic idea of (12) can be verified using the connectivity of the graph, which can be expressed as

$$\forall v_i, v_j : \varepsilon_{ik_1} \times \varepsilon_{k_1k_2} \times \cdots \times \varepsilon_{k_\mu j} \neq 0 \quad (13)$$

where ε_{ik_1} denote the edge weight between nodes v_i and v_{k_1}. The node v_i can learn the more diverse features from its neighbor node v_j when μ is larger. Moreover, any two nodes belonging to neighbors through the constraint of (12) are likely to benefit from each other. In fact, the key connectivity restriction in (12) not only preserves the label of surrounding neighbors to be consistent but also preserves the diversity of any two nodes belonging to the same category achieved by underlying μ neighbors. When the weight value of interclass connections of graph structure is large enough to be the same as the intraclass weight, the threshold method such as (10) will be failed, and the large weight values of interclass connections can be removed by the strength of underlying neighbors. Although the weight of interclass connections is large, their underlying neighbor connections are probably none because the pixels belonging to same class are more likely equipped with more potential neighbors. Consequently, the large weight values of interclass connections can be effectively removed and the diversity of intraclass information can be effectively preserved to a certain degree via underlying μ neighbors. When the distributions of surrounding neighbors are different and the labels of surrounding neighbors are consistent, the GCN will aggregate beneficial information from surrounding neighbors of intraclass pixels. Consequently, the key connectivity of diverse neighbors forces them into the same class and eliminates the ambiguous connections of the hard samples.

The implementation of (12) is an NP-hard problem. To demonstrate its significance, the process is instantiated by setting $\mu = 1$, indicating that the path between nodes v_i and v_j must contain a different neighbor. Afterward, the adjacency similarity matrix $\bar{\mathbf{W}}$ constrained by (12) can be rewritten as

$$\begin{aligned}\hat{\mathbf{W}} &= f(\bar{\mathbf{W}}) \\ &= \Phi\left(v_i, \mathbb{N}_{\lambda_a}(v_k)\right) \times \Phi\left(v_k, \mathbb{N}_{\lambda_a}(v_j)\right) \times \Phi\left(v_j, \mathbb{N}_{\lambda_a}(v_i)\right) \\ &= 1. \end{aligned} \quad (14)$$

Finally, the new adjacency matrix $\hat{\mathbf{W}} = f(\bar{\mathbf{W}})$ can be obtained by (14). The given graph $\mathbf{G}$ can be optimized by equations (10) and (14), and the final adjacency matrix $\hat{\mathbf{W}}$ can be obtained to guarantee the quality of diverse neighbors. As the basis of (14) and (10) aims to obtain the diverse neighbors from the topological extension and remove weak interclass connections between neighbors coarsely in real HSI. Equation (14) aims to remove strong interclass connections finely based on the diverse neighbors. Therefore, the order of the optimization algorithm cannot be changed. Through these two critical steps, the plug-and-play post-processing framework that is independent of the network architecture is formulated.

The proposed DCGCN method is a general post-processing module that can be used as a complement to GCN-based algorithms. The main procedures of the DCGCN algorithm based on a given graph for HSI classification are summarized in Algorithm 1.

Algorithm 1 DCGCN Method for HSI Classification

Input: Given the HSI feature matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N] \in \mathbb{R}^{N \times b}$ and an initial graph $\mathbf{G}$ from traditional GCN-based methods.

1) Obtain the adaptive smoothing matrix $\bar{\mathbf{W}}$ via equation (10) to guarantee the diversity quality of the surrounding neighbors by using a faithful graph-specific parameter λ_a from equation (11).

2) Obtain the final adjacency matrix $\hat{\mathbf{W}}$ via equation (14) to guarantee to correct the connections of the hard samples by using the consecutive one neighbor.

3) Perform graph convolution $\hat{\mathbf{D}}^{-1/2}\hat{\mathbf{W}}\hat{\mathbf{D}}^{-1/2}\mathbf{H}^{(k-1)}$ on the previous feature matrix $\mathbf{H}^{(k-1)}$ from k-1 layer, $\mathbf{H}^{(0)} = \mathbf{X}$.

4) Perform the GCN method to aggregate the beneficial information from diverse and connected neighbors.

Output: Classification results for test samples.

IV. Theoretical Analysis

In this section, the theoretical spectral analysis of the proposed DCGCN method is introduced. The DCGCN method corresponds to a fixed filter on the spectral graph domain. In addition, DCGCN acts as a low-pass filter to produce smooth features on the graph, facilitating nearby nodes to share similar representations or share the same labels.

A. Spectral Graph Convolution

The spectral convolution of the GCN model is based on graph Fourier analysis, which relies on the spectral decomposition of graph Laplacians. The graph Laplacian can be defined as $\mathbf{L} = \mathbf{D} - \mathbf{A}$, and it can be normalized as $\mathbf{L}_{\text{sym}} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$. $\mathbf{L}$ is a symmetric positive semidefinite matrix with eigen decomposition $\mathbf{L} = \mathbf{U}\Lambda\mathbf{U}^{\mathrm{T}}$, where $\mathbf{U} \in \mathbb{R}^{N \times N}$ denotes orthonormal eigenvectors and $\Lambda \in \mathbb{R}^{N \times N} = \text{diag}(\lambda_1, \ldots, \lambda_N)$ denotes the diagonal matrix with eigenvalues. The graph Fourier transform can be defined as the eigen decomposition of the graph Laplacian, where eigenvectors are Fourier bases and eigenvalues represent frequencies of the graph. Given a signal $\mathbf{X} \in \mathbb{R}^N$ (scalar of each node) defined

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on the vertices of the graph, the graph Fourier transform of $\mathbf{X}$ is $\hat{\mathbf{X}} = \mathbf{U}^{\mathrm{T}}\mathbf{X}$ and the inverse operation is $\mathbf{X} = \mathbf{U}\hat{\mathbf{X}}$. Therefore, the graph convolution operation between the given signal $\mathbf{X}$ and the filter $\mathbf{F}$ can be defined as

$$\mathbf{F} \times \mathbf{X} = \mathbf{U}\left(\left(\mathbf{U}^{\mathrm{T}}\mathbf{F}\right) \odot \left(\mathbf{U}^{\mathrm{T}}\mathbf{X}\right)\right) = \mathbf{U}\hat{\mathbf{F}}\mathbf{U}^{\mathrm{T}}\mathbf{X} \tag{15}$$

where $\mathbf{F} = \mathrm{diag}(\hat{\mathbf{f}}_1, \ldots, \hat{\mathbf{f}}_N)$ denotes a diagonal matrix with spectral filter coefficients. The basic idea of (15) is that convolution in the time domain corresponds to an inverse Fourier transform of multiplication in the Fourier domain [52]. Bruna et al. [37] approximated (15) by using rth order polynomials of graph Laplacians

$$\mathbf{U}\hat{\mathbf{F}}\mathbf{U}^{\mathrm{T}}\mathbf{X} \approx \sum_{i=0}^{r} \theta_i \mathbf{L}^i \mathbf{X} = \mathbf{U}\left(\sum_{i=0}^{r} \theta_i \Lambda^i\right)\mathbf{U}^{\mathrm{T}}\mathbf{X} \tag{16}$$

where θ_i denotes filter coefficients corresponding to polynomials of the Laplacian eigenvalues, i.e., $\hat{\mathbf{F}} = \sum_i \theta_i \Lambda^i$ or $\hat{f}(\lambda_j) = \sum_i \theta_i \lambda_j^i$. Considering the large computation in (16), Defferrard et al. [36] proposed to use Chebyshev polynomials to approximate graph convolution, and Chebyshev polynomials are recursively given by $T_i(x) = 2xT_{i-1}(x) - T_{i-2}(x)$ with $T_0(x) = 1$ and $T_1(x) = x$. The filtering operation can be expressed as

$$\mathbf{U}\hat{\mathbf{F}}\mathbf{U}^{\mathrm{T}}\mathbf{X} = \sum_{i=0}^{r} \theta_i T_i\left(\mathbf{L}'_{\mathrm{sym}}\right)\mathbf{X} \tag{17}$$

where $\mathbf{L}'_{\mathrm{sym}} = 2/\lambda_{\max}\mathbf{L}_{\mathrm{sym}} - \mathbf{I}$ denotes the scaled normalized Laplacian to force the eigenvalue to lie between -1 and 1. Furthermore, to address the overfitting problem of local neighbor structures for wide node degree distributions in [36], Kipf and Welling [38] used the affine approximation of $r = 1$ and $\lambda_{\max} = 2$ to define GCNs. In practice, it can be in favor of constraining the number of parameters for further addressing overfitting. Further, by setting the coefficients $\theta = \theta_0 = -\theta_1$, the basic GCN can be defined as follows:

$$\mathbf{F} \times \mathbf{X} = \theta\left(\mathbf{I} + \bar{\mathbf{D}}^{-1/2}\bar{\mathbf{A}}\bar{\mathbf{D}}^{-1/2}\right)\mathbf{X} \tag{18}$$

where $\bar{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ denotes the renormalization trick.

B. Low-Pass Filtering

In practice, to handle potential issues associated with the first-order Chebyshev filter, Kipf and Welling [38] proposed the renormalization trick, which adds self-loops for all nodes. The augmented normalized adjacency matrix can be expressed as $\mathbf{S}_{\mathrm{GCN}} = \bar{\mathbf{D}}^{-1/2}\bar{\mathbf{A}}\bar{\mathbf{D}}^{-1/2}$. Correspondingly, the augmented normalized Laplacian is $\bar{\mathbf{L}}_{\mathrm{sym}} = \mathbf{I} - \bar{\mathbf{D}}^{-1/2}\bar{\mathbf{A}}\bar{\mathbf{D}}^{-1/2}$, and $\mathbf{S}_{\mathrm{GCN}} = \mathbf{I} - \bar{\mathbf{L}}_{\mathrm{sym}} = \mathbf{U}(\mathbf{I} - 1/2\Lambda)\mathbf{U}^{\mathrm{T}}$. Thus, the spectral filter of the GCN model can be described as a polynomial of eigenvalues of the underlying Laplacian, i.e., $\hat{f}(\bar{\lambda}_i) = (1 - \bar{\lambda}_i)^k$, where $\bar{\lambda}_i$ are eigenvalues of $\bar{\mathbf{L}}_{\mathrm{sym}}$.

The graph structure of the proposed DCGCN method is a modified version of the GCN model, which can be obtained by topological extension with (8) and then by constraints of the diversity and the connectivity based on (10) and (14). By adopting the Taylor approximation of the inverse operation $(\mathbf{I} + \mathbf{L}/\eta)^{-1}$ in (8), which can be decomposed as

$$\left(\mathbf{I} + \mathbf{L}/\eta\right)^{-1} = \mathbf{I} - \left(\frac{1}{\eta}\right)\mathbf{L} + \cdots + \left(-\frac{1}{\eta}\right)^n \mathbf{L}^n + \cdots \tag{19}$$

The first-order truncated form of (19) is

$$\left(\mathbf{I} + \mathbf{L}/\eta\right)^{-1} \approx \mathbf{I} - \left(\frac{1}{\eta}\right)\mathbf{L}. \tag{20}$$

The topological extension $\mathbf{W}$ can be represented by the first-order approximation, which can be rewritten as

$$F(\mathbf{W}) = \mathbf{I} - \left(\frac{1}{\eta}\right)\mathbf{L} \tag{21}$$

where $F(\mathbf{W})$ denotes the first-order approximation of the Taylor. Therefore, the obtained adjacency matrix through the proposed DCGCN method can be expressed as

$$F\left(\hat{\mathbf{W}}\right) = \mathbf{I} - \left(\frac{1}{\eta}\right)\hat{\mathbf{L}}_{\mathrm{sym}}. \tag{22}$$

Correspondingly, the spectral filter of the DCGCN model can be described as $\hat{f}(\hat{\lambda}_i) = (1 - (1/\eta)\hat{\lambda}_i)^k$, where $\hat{\lambda}_i$ are eigenvalues of $\hat{\mathbf{L}}_{\mathrm{sym}}$ based on the final graph $\hat{\mathbf{W}}$. Based on the above analysis, the spectral filter of the GCN model is a special case of the proposed DCGCN. Fig. 4 demonstrates that the low-pass property of the DCGCN model is better than that of the GCN model because the DCGCN model preserves smaller eigenvalues and eliminates irrelevant high-frequency information. It can be seen that when eigenvalues are large, there is no frequent response. Here, the parameter η is set by two empirically. The spectral filter of the DCGCN model then becomes $\hat{f}(\hat{\lambda}_i) = (1 - (1/2)\hat{\lambda}_i)^k$, which acts as a low-pass filter compared with the GCN model $\hat{f}(\bar{\lambda}_i) = (1 - \bar{\lambda}_i)^k$.

V. Experimental Analysis

This section presents experimental results on three real HSI datasets, including, Indian Pines, Houston University, and HyRANK. Moreover, the basic ideas of the proposed DCGCN are deployed into several popular GCN-based HSI classification methods to test their effectiveness. These GCN-based methods are GCN [38], S^{2}GCN [25], MDGCN [24], FuNet-C [30], CEGCN [29], and GCGCN [26] methods, respectively.

A. Dataset Description

Three real HSI datasets are introduced as follows.

1) The Indian Pines image was acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor during a flight campaign on June 12, 1992, over Northwestern, Indiana, USA, with a size of 145 × 145 with 220 spectral channels. After removing 20 bands corrupted by the atmospheric water absorption effect, the remaining 200 spectral channels have been considered for the experiments. Moreover, the spatial resolution was 20 m and the spectral resolution was 10 nm, covering a spectral range of 200–2400 nm. This scene covered an agricultural field and contained 16 classes. The false-color image and the ground truth were shown in Fig. 5(a) and (b).

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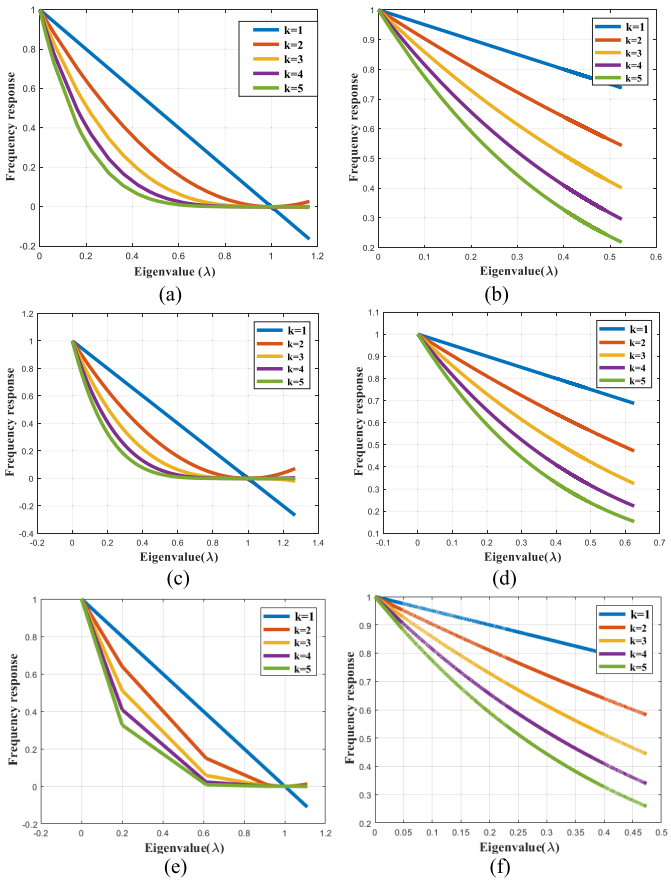


Fig. 4. Frequency response between GCN model and DCGCN model on three datasets. (a) and (b) Indian Pines. (c) and (d) Houston. (e) and (f) HyRANK.

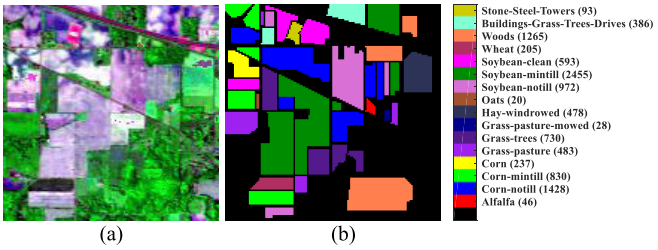


Fig. 5. Indian Pines data. (a) False color image (bands 20, 69, and 178 are red, green, and blue, respectively). (b) Ground truth image.

2) The Houston University (Houston) 2013 dataset was captured by the compact airborne spectrographic imager (CASI) over the University of Houston campus and the neighboring urban area in June 2013. The Houston image contains 349 × 1905 pixels with 144 spectral bands. In addition, the spatial resolution was 2.5 m. The false-color and the corresponding ground-truth of Houston were shown in Fig. 6(a) and (b).
3) The HyRANK dataset, namely Dioni, was from the Hyperion sensor (EO-1, USGS). The Dioni dataset contains 250 × 1376 pixels with 176 spectral bands, which contains 12 land-cover types in the dataset. The false color and corresponding ground-truth images of Dioni were shown in Fig. 7(a) and (b).

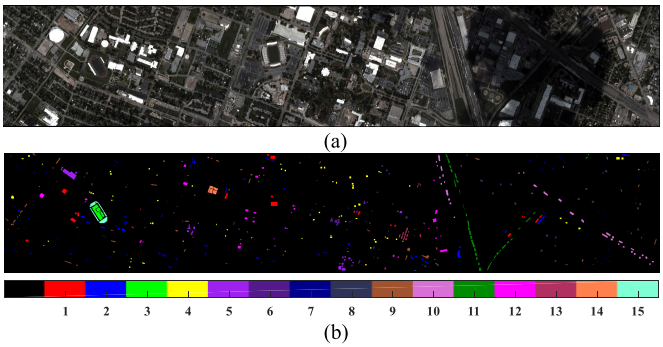


Fig. 6. Houston data. (a) False color image (bands 45, 38, and 32 are red, green, and blue, respectively). (b) Ground truth image.

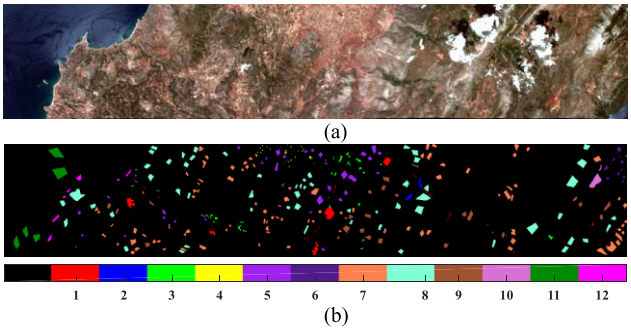


Fig. 7. HyRANK data with Dioni. (a) False color image (bands 23, 11, and 7 are red, green, and blue, respectively). (b) Ground truth image.

B. Experimental Evaluation Metrics

It has been a common practice in hyperspectral classification that per-class accuracy (CA), overall accuracy (OA), average accuracy (AA), and kappa coefficient (KC) are major measures to be used to evaluate classification performance. Consider an M-class confusion matrix constructed in Table I. Based on Table I, specific explanations with CA, OA, AA, and KC are introduced as follows. M denotes the number of classes, n_{ji} denotes the number of samples in the ith class that have been classified in the jth class, C_i denotes the ground truth, $\hat{C}_i$ denotes the prediction result, and N denotes the total number of HSI pixels, $N = \sum_{i=1}^{M} n_i$. The following classification measures are used to assess the classification performance:

1) CA is defined as

$$\mathrm{CA}(C_i) = \frac{n_{ii}}{\sum_{j=1}^{M} n_{ji}}. \tag{23}$$

2) OA is defined as

$$\mathrm{OA} = \frac{1}{N} \sum_{i=1}^{M} n_{ii}. \tag{24}$$

3) AA is defined as

$$\mathrm{AA} = \frac{1}{M} \sum_{i=1}^{M} CA(C_i). \tag{25}$$

4) KC is defined as

$$\mathrm{KC} = \frac{p_o - p_e}{1 - p_e} \tag{26}$$

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TABLE I
M-CLASS CONFUSION MATRIX

Truth / Prediction	C_1	C_2	...	C_M
$\hat{C}_1$	n_{11}	n_{12}	...	n_{1M}
$\hat{C}_2$	n_{21}	n_{22}	...	n_{2M}
...	...	...	...	...
$\hat{C}_M$	n_{M1}	...	...	n_{MM}

where $p_o = ((\sum_{i=1}^{M} n_{ii})/N)$ denotes the sum of the number of correctly classified samples in each category divided by the total number of samples, $p_e = (((\sum_{j=1}^{M} n_{ij}) \times (\sum_{i=1}^{M} n_{ij}))/N \times N)$ denotes that the actual number of samples in each category is multiplied by the predicted number of samples in each category, and then divided by the total number of samples.

C. Experimental Results

Table II shows the number of training and testing samples in the experiments. Tables III–V shows the classification results of different GCN-based methods on three real HSI datasets. Moreover, the basic idea of the diversity-connected graph was deployed in other GCN-based methods to validate the effectiveness of the diversity-connected graph. Overall, the results show that all the GCN-based methods with the prefix diversity and connectivity (DC)-are improved methods with a quality graph. From Tables III–V, it can be observed that all GCN-based methods with DC-prefixes improve better than original GCN-based in three common indices of OA, AA, and KC. Notably, the numbers reported in bold type indicate the best results compared to the original classification methods. The diversity-connected strategy can improve the quality of the graph structure in GCN architectures. Based on the above results, several conclusions can be derived.

From Table III, the traditional GCN method has poor classification performance because the simple graph structure relies only on single spectral information and cannot encode these complex spectral-spatial signatures. The classification performance of the GCN method is below 60% in the Indian Pines data because the spectra of intraclass samples are highly variable. Then, the improvement with GCN is to construct the novel diverse and connectivity graph by removing the noisy connections due to spectral variability of intraclass pixels. The novel graph from DCGCN method obtain the better classification performance compared to GCN method achieving an improvement of approximately 6%, 2%, and 6% with OA, AA, and KC. The S^2GCN, FuNet-C, MDGCN, CEGCN, and GCGCN methods construct the spectral-spatial graph obtaining better results compared the classification performance of GCN method. Interestingly, the DCS^{2}GCN, DCFuNet-C, DCMDGCN, DCCEGCN, and DCGCGCN methods can achieve effective improvement with OA, AA, and KC by using DC restrictions. From Table IV, the MDGCN method uses the dynamic graph structure to encode complex spectral-spatial features, showing better classification performance on Houston datasets. The DCMDGCN method obtain the better classification performance compared to MDGCN method achieving an improvement of approximately 3% with OA. The GCGCN method uses the global graph structure to encode complex spectral-spatial features, showing better classification performance on Houston datasets. The DCGCGCN method obtain the better classification performance compared to GCGCN method achieving an improvement of approximately 2% with OA, AA, and KC. This phenomenon demonstrates that the basic idea of diverse-connected graph not only improve the quality of complex graph structure but also improve the quality of the above local graph structure such as GCN method. From Table V, the FuNet-C and CEGCN methods achieve good classification performance on the HyRANK dataset by integrating the CNN and GCN architectures. The DCFuNet-C and DCCEGCN methods also achieving better classification results with improvements on OA, AA, and KC. This phenomenon demonstrates that considering the topological restrictions of the DC is also beneficial to improve the initial graph structure. In addition, as reflected by visual results in Figs. 8–10, all the GCN-based methods with the prefix DC- are also greatly improved.

D. Analysis of Different Training Samples

To further test the effectiveness of the basic idea of the proposed DCGCN, a series of experiments between GCN-based methods and improved GCN-based methods with the prefix DC- are performed on three datasets under different training samples. The number of training samples selected for each class ranges from 5 to 30, with an interval of five. Experimental results demonstrate that the OA of the improved GCN-based methods with the prefix DC- under different training samples is higher than that of original GCN-based methods. From Fig. 11, it can be seen that the performance of the proposed DC graph is remarkable when the number of training samples is very limited. The main reason for this phenomenon is that the proposed DC graph can obtain a more homogeneous graph by preserving the connectivity of intraclass samples and removing the incorrected relationship of interclass samples. Because of the large spectral variation of intraclass samples, it is a challenge to construct a homogenous graph for GCN-based methods when the number of training samples is very limited.

E. Analysis of Ablation Study

As mentioned in Section III, the proposed DCGCN contains two critical parts (the diversity and the connectivity constraints) to improve the HSI classification performance. In this study, three datasets, (i.e., Indian Pines, Houston 2013, and HyRANK) are used to demonstrate the usefulness of these two constraints. The number of training samples per class is five. Table VI shows the corresponding classification results using two constraints to illustrate the importance of these two constraints. DGCN denotes that the GCN method with the diversity restriction. It can be observed

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TABLE II

DETAILED NUMBER OF TRAINING AND TESTING SAMPLES ON INDIAN PINES, HOUSTON, AND HYRANK DATASETS

	Indian Pines			Houston				HyRANK		
Class	Name	Train	Test	Name	Train	Test	Class	Name	Train	Test
1	Alfalfa	5	41	Healthy grass	5	1246	1	Dense Urban Fabric	5	1257
2	Corn-notill	5	1423	Stressed grass	5	1249	2	Mineral Extraction Sites	5	199
3	Corn-mintill	5	825	Synthetic grass	5	692	3	Non Irrigated Arable Land	5	609
4	Corn	5	232	Tree	5	1239	4	Fruit Trees	5	145
5	Grass-pasture	5	478	Soil	5	1237	5	Olive Groves	5	1763
6	Grass-trees	5	725	Water	5	320	6	Coniferous Forest	5	356
7	Grass-pasture-mowed	5	23	Residential	5	1263	7	Dense Sclerophyllous Vegetation	5	5030
8	Hay-windrowed	5	473	Commercial	5	1239	8	Sparce Sclerophyllour Vegetation	5	6369
9	Oats	5	15	Road	5	1247	9	Sparcely Vegetated Areas	5	1749
10	Soybean-notill	5	967	Highway	5	1222	10	Rock and Sand	5	487
11	Soybean-mintill	5	2450	Railway	5	1230	11	Water	5	1607
12	Soybean-clean	5	588	Parking lot 1	5	1228	12	Coastal Water	5	393
13	Wheat	5	200	Parking lot 2	5	464		Total	60	19964
14	Woods	5	1260	Tennis court	5	423				
15	Buildings-Grass-Trees-Drives	5	381	Running track	5	655				
16	Stone-Steel-Towers	5	88	Total	75	14954				
	Total	80	10619							

TABLE III

CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS VIA DCGCN ALGORITHM ON INDIAN PINES DATASET

Class	GCN	DCGCN	S^2GCN	DCS^{2}GCN	FuNet-C	DCFuNet-C	MDGCN	DCMDGCN	CEGCN	DCCEGCN	GCGCN	DCGCGCN
1	100(0)	94.63(3.18)	90.65(15.15)	82.93(8.90)	82.11(11.81)	78.05(15.73)	95.12(4.22)	91.06(1.99)	87.31(3.23)	**92.68**(10.66)	95.53(1.00)	**97.56**(0)
2	31.23(16.58)	**42.22**(12.14)	43.05(14.31)	**47.10**(13.61)	37.08(13.37)	33.95(10.93)	58.59(12.92)	**66.23**(8.80)	44.95(11.99)	40.75(15.96)	50.96(16.78)	**60.81**(8.72)
3	18.55(18.26)	**52.19**(10.13)	57.56(13.53)	**67.74**(10.41)	31.47(7.26)	**39.52**(6.11)	69.09(22.94)	66.30(10.69)	48.04(16.48)	47.07(18.94)	71.94(19.23)	**72.41**(23.31)
4	97.49(6.16)	79.48(7.82)	83.48(15.41)	**84.77**(11.08)	43.32(11.93)	**52.66**(10.17)	96.98(0.75)	87.93(14.75)	50.03(13.87)	**56.09**(16.39)	91.52(10.99)	**95.95**(8.59)
5	67.78(15.58)	**70.71**(10.51)	68.52(10.52)	**79.01**(10.28)	66.88(8.31)	65.62(11.56)	77.62(10.51)	**82.11**(5.37)	75.82(6.94)	75.79(10.46)	83.65(15.55)	**89.46**(6.71)
6	61.77(6.91)	**76.64**(19.32)	92.30(5.81)	84.32(18.35)	66.16(9.88)	**71.79**(7.15)	85.15(9.10)	**87.03**(4.90)	97.51(2.89)	95.85(2.27)	88.99(15.71)	**91.36**(5.90)
7	99.28(1.78)	96.52(3.64)	100(0)	**100**(0)	91.30(7.78)	**94.20**(4.49)	100(0)	**100**(0)	100(0)	**100**(0)	98.55(2.25)	**100**(0)
8	100(0)	99.92(0.19)	75.94(12.21)	**96.69**(4.96)	68.96(20.76)	**79.53**(14.59)	99.08(1.59)	**100**(0)	90.63(6.67)	**92.97**(5.98)	94.43(13.64)	**100**(0)
9	100(0)	**100**(0)	100(0)	**100**(0)	100(0)	96.67(5.58)	100(0)	**100**(0)	100(0)	**100**(2.14)	92.22(7.79)	**100**(0)
10	53.09(19.95)	**72.58**(7.08)	57.45(7.02)	**65.12**(16.27)	38.52(6.87)	36.37(10.24)	67.91(8.21)	**72.70**(9.24)	67.20(22.29)	61.25(12.95)	72.51(15.15)	**80.12**(6.86)
11	52.35(20.72)	**59.97**(5.02)	48.01(15.08)	**63.01**(7.65)	41.47(12.83)	**49.79**(16.92)	60.05(18.47)	**66.20**(19.49)	57.84(16.14)	**78.88**(14.73)	73.99(20.94)	**74.96**(8.24)
12	42.43(8.85)	**51.77**(5.76)	60.86(23.12)	**72.45**(17.49)	37.13(9.92)	33.70(6.87)	62.98(18.94)	60.49(12.04)	59.97(7.92)	45.02(11.55)	73.90(13.98)	67.55(8.36)
13	97.33(5.81)	**98.80**(1.15)	97.08(3.84)	95.00(2.66)	96.83(5.56)	**97.25**(3.90)	99.17(1.44)	98.33(1.29)	100(0.15)	99.90(0.20)	97.25(1.44)	**99.00**(0)
14	88.94(6.49)	64.76(13.81)	88.39(9.88)	83.45(9.70)	64.38(8.26)	61.96(17.29)	90.98(7.19)	90.42(5.34)	84.59(7.15)	**88.61**(10.81)	99.05(0.86)	89.52(6.22)
15	76.90(0.17)	69.45(15.36)	83.64(11.83)	**92.08**(4.16)	38.71(6.20)	36.22(7.19)	74.72(22.75)	**75.68**(17.14)	89.29(8.14)	87.08(21.15)	84.12(7.78)	**84.41**(21.54)
16	99.05(0.86)	95.45(4.17)	96.21(3.57)	**96.21**(4.75)	87.38(3.02)	**94.51**(5.77)	96.97(2.62)	96.02(1.39)	99.77(0.45)	99.31(0.45)	99.24(0.59)	97.96(0.51)
OA	57.90(8.60)	**63.88**(4.96)	64.05(2.46)	**71.00**(4.11)	48.38(2.91)	**50.90**(2.40)	72.53(2.22)	**75.28**(4.16)	67.23(4.20)	**70.76**(3.00)	77.66(6.92)	**79.35**(1.37)
AA	74.14(4.13)	**76.57**(2.98)	77.70(2.98)	**81.87**(2.62)	58.37(1.93)	**60.10**(1.44)	83.40(0.83)	**83.78**(2.23)	78.31(2.72)	**78.82**(2.11)	85.49(5.24)	**87.57**(1.18)
KC	53.78(8.85)	**59.57**(5.47)	59.87(2.41)	**67.43**(4.52)	42.23(3.08)	**44.85**(2.32)	69.26(2.38)	**72.14**(4.60)	63.14(4.62)	**66.46**(3.27)	74.69(7.87)	**76.63**(1.53)

TABLE IV

CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS VIA DCGCN ALGORITHM ON HOUTON 2013 DATASET

Class	GCN	DCGCN	S^2GCN	DCS^{2}GCN	FuNet-C	DCFuNet-C	MDGCN	DCMDGCN	CEGCN	DCCEGCN	GCGCN	DCGCGCN
1	86.89(6.23)	85.61(4.46)	90.71(6.42)	86.60(2.53)	89.17(8.04)	86.32(4.92)	86.56(9.03)	80.31(4.27)	97.34(5.70)	**97.94**(5.31)	89.90(6.28)	**91.72**(6.22)
2	77.85(16.59)	**90.54**(8.12)	88.58(14.28)	**94.42**(5.39)	87.40(10.30)	83.01(8.77)	75.34(7.41)	**81.55**(2.73)	87.83(13.02)	**91.24**(14.01)	85.86(2.18)	**92.00**(5.93)
3	96.12(2.69)	**97.57**(1.28)	97.11(6.10)	**98.65**(0.95)	95.04(2.62)	93.64(4.77)	97.75(1.38)	96.29(4.50)	99.94(2.99)	**100**(0.23)	99.07(0.08)	**99.55**(0.21)
4	90.20(2.80)	87.33(2.28)	94.57(3.71)	**95.45**(2.39)	93.52(3.67)	86.09(10.65)	90.89(4.15)	85.40(8.66)	97.63(3.49)	**98.23**(3.99)	96.66(3.25)	95.89(2.89)
5	91.39(8.38)	**93.80**(5.29)	95.82(7.05)	91.04(10.26)	96.12(4.37)	94.81(4.12)	94.23(9.47)	92.87(5.81)	99.96(4.35)	**100**(5.70)	99.42(0.24)	97.98(2.72)
6	87.66(6.37)	84.43(5.30)	83.44(5.78)	**85.39**(9.87)	86.35(4.98)	83.33(7.12)	80.73(8.58)	79.43(5.61)	91.85(6.67)	**94.11**(3.23)	92.25(6.41)	89.56(5.67)
7	59.11(14.59)	**67.72**(12.95)	59.53(15.19)	**67.09**(13.35)	39.85(11.73)	**57.09**(5.90)	75.50(7.93)	**80.48**(9.09)	92.68(13.47)	**97.86**(11.50)	75.58(6.27)	73.63(11.87)
8	48.54(10.32)	43.44(11.89)	43.01(9.12)	42.98(14.40)	57.25(17.50)	46.97(12.45)	45.56(10.33)	45.17(8.13)	66.67(12.52)	59.52(13.02)	44.99(5.26)	**52.81**(12.36)
9	49.93(17.96)	**66.22**(10.81)	67.32(4.43)	**71.56**(5.71)	61.75(11.04)	51.24(12.21)	67.42(9.80)	**78.13**(5.35)	86.50(15.22)	83.13(10.10)	71.52(11.69)	67.96(9.56)
10	55.82(10.85)	**60.79**(12.18)	62.25(10.91)	**75.25**(12.38)	51.34(13.76)	**58.25**(9.89)	44.05(15.24)	**52.95**(13.25)	**100**(16.68)	**100**(17.11)	75.25(17.77)	**78.26**(25.58)
11	47.18(16.75)	**51.00**(15.85)	66.79(12.02)	66.39(10.21)	50.84(23.97)	**57.99**(9.61)	56.45(2.66)	**60.33**(14.36)	**100**(13.28)	**100**(11.84)	80.65(8.68)	**84.64**(10.61)
12	40.81(14.23)	**41.63**(9.85)	59.69(11.57)	58.27(15.60)	46.47(1.87)	**58.01**(11.95)	49.23(24.73)	**50.16**(10.64)	59.53(13.50)	**72.12**(12.34)	49.22(6.98)	**68.40**(7.41)
13	27.44(6.99)	**29.24**(4.87)	75.40(5.65)	**77.21**(11.44)	40.16(6.92)	**48.71**(14.74)	69.91(11.45)	58.17(16.43)	59.66(20.26)	57.46(17.40)	53.54(17.50)	52.07(9.38)
14	95.47(2.37)	93.38(4.42)	**100**(0)	**100**(0)	79.12(9.82)	**90.27**(5.22)	97.76(1.21)	**98.63**(0.93)	99.86(0)	**100**(0)	98.72(2.60)	**100**(0)
15	98.40(0.54)	97.99(1.51)	**100**(0)	**100**(0)	98.98(0.64)	**97.33**(2.58)	91.85(1.29)	91.66(9.17)	99.45(0.18)	98.78(4.09)	99.72(0.07)	99.15(2.22)
OA	67.93(2.20)	**71.28**(0.91)	76.23(1.30)	**78.13**(2.98)	69.95(0.73)	**70.83**(2.94)	71.55(0.58)	**73.15**(2.85)	88.36(2.82)	**89.92**(3.14)	79.18(1.16)	**81.92**(1.21)
AA	70.19(2.04)	**72.71**(0.95)	78.95(1.43)	**80.69**(2.12)	67.08(0.68)	**68.32**(2.57)	74.88(0.41)	**75.44**(2.63)	89.26(3.05)	**90.03**(2.87)	80.83(1.33)	**82.91**(0.92)
KC	65.40(2.37)	**69.00**(0.96)	74.36(1.40)	**76.41**(3.19)	67.56(0.79)	**68.50**(3.19)	69.31(0.55)	**70.97**(3.11)	87.35(3.60)	**89.04**(3.41)	77.53(1.27)	**80.47**(1.32)

from Table VI that the DGCN and DCGCN methods outperform the GCN method on three datasets in terms of OA, AA, and KC, indicating the effectiveness of incorporating these two constraints into the graph structure. The main reason for this phenomenon is that the diversity constraint captures the underlying beneficial neighbors according to Theorem 1, and the connectivity preserves critical relationships of intraclass samples to further refine the graph structure.

F. Parameter Analysis

There are mainly three parameters in this manuscript including parameters λ_a, μ, and η. The parameter λ_a controls the diversity quality of surrounding neighbors in removing weak interclass connections of graph structure, which can be adaptively obtained by available training samples. The parameter μ controls the strength of underlying connectivity in removing the strong interclass connections of graph structure,

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TABLE V
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS VIA DCGCN ALGORITHM ON HYRANK DATASET

Class	GCN	DCGCN	S^2GCN	DCS^{2}GCN	FuNet-C	DCFuNet-C	MDGCN	DCMDGCN	CEGCN	DCCEGCN	GCGCN	DCGCGCN
1	79.41(5.72)	70.30(8.62)	77.42(13.03)	75.35(10.07)	47.16(11.93)	**53.59**(3.77)	72.91(18.17)	**74.17**(5.77)	47.67(23.06)	**65.09**(10.35)	72.46(10.91)	71.29(14.03)
2	62.98(18.60)	**80.91**(1.42)	92.72(9.53)	89.70(12.46)	78.52(2.33)	**78.56**(1.76)	84.49(14.19)	**97.29**(6.83)	94.77(14.34)	**99.40**(6.84)	90.62(7.50)	87.77(8.10)
3	39.68(19.43)	**41.96**(34.49)	69.27(12.36)	**72.71**(11.91)	69.21(13.08)	64.53(12.82)	76.14(13.26)	**90.57**(5.27)	75.88(4.90)	**79.67**(9.10)	70.25(9.50)	**76.63**(6.86)
4	84.14(10.02)	**91.38**(4.38)	83.68(7.77)	81.15(7.23)	91.03(3.47)	87.82(3.54)	87.00(6.14)	**92.00**(3.84)	60.28(21.02)	**72.55**(18.22)	86.44(14.23)	**88.28**(7.62)
5	36.09(8.01)	**43.93**(1.48)	76.37(10.81)	72.12(10.42)	65.10(7.81)	**74.83**(3.57)	76.06(11.38)	75.20(12.01)	76.01(9.42)	**77.32**(8.87)	67.39(6.14)	**76.04**(5.48)
6	96.63(4.46)	90.59(1.39)	100(0)	96.21(4.56)	97.82(0.93)	**98.97**(0.86)	**100(0)**	**100**(0)	**100**(0)	**100**(0)	99.95(0.11)	99.91(0.14)
7	76.97(7.60)	70.93(2.28)	79.30(9.42)	**81.49**(5.67)	84.01(7.10)	**85.35**(1.84)	85.42(3.98)	84.00(7.66)	88.40(8.64)	86.71(7.48)	83.02(6.56)	82.72(4.41)
8	59.62(19.19)	**73.65**(8.29)	74.23(13.13)	**80.63**(9.04)	71.29(14.77)	70.31(15.59)	71.48(11.75)	**72.18**(14.73)	63.89(16.37)	**66.18**(16.87)	84.65(15.53)	**89.80**(5.19)
9	44.77(19.43)	41.23(8.41)	63.60(11.72)	**68.13**(20.44)	46.88(10.75)	45.66(15.08)	48.87(7.41)	**66.73**(8.28)	65.44(17.38)	**68.91**(14.08)	70.69(19.01)	**71.19**(13.89)
10	89.18(1.57)	87.47(4.36)	93.02(3.35)	92.13(4.38)	94.25(4.16)	**94.73**(4.11)	94.69(4.58)	87.93(4.62)	97.37(3.65)	97.33(3.74)	95.00(4.28)	90.66(8.40)
11	70.94(10.10)	**72.13**(0.62)	98.40(1.92)	91.24(7.44)	97.74(2.89)	97.32(1.40)	88.58(11.05)	**99.91**(0.06)	**100**(0)	**100**(0)	96.07(3.23)	**97.29**(4.23)
12	95.25(3.75)	**96.06**(4.86)	82.83(16.55)	**86.98**(17.08)	67.43(22.53)	67.35(24.61)	92.77(3.30)	**95.95**(12.79)	99.54(8.10)	99.39(12.50)	85.71(4.81)	79.26(10.10)
OA	64.44(2.39)	**67.48**(2.98)	78.10(3.05)	**80.06**(3.21)	73.53(4.43)	**74.54**(4.65)	76.76(4.79)	**79.61**(4.97)	75.96(4.52)	**78.04**(4.95)	81.82(6.13)	**84.17**(3.24)
AA	69.64(0.63)	**71.71**(1.92)	82.32(2.79)	**82.57**(3.05)	70.04(3.26)	**70.69**(3.40)	81.53(2.71)	**86.33**(1.92)	80.77(1.84)	**84.38**(2.47)	83.52(2.44)	**84.24**(3.62)
KC	57.69(2.34)	**60.96**(3.22)	73.63(3.43)	**75.76**(3.70)	67.80(4.96)	**68.98**(5.23)	71.85(5.38)	**75.23**(5.78)	70.79(5.12)	**73.34**(5.58)	77.73(7.14)	**80.46**(3.90)

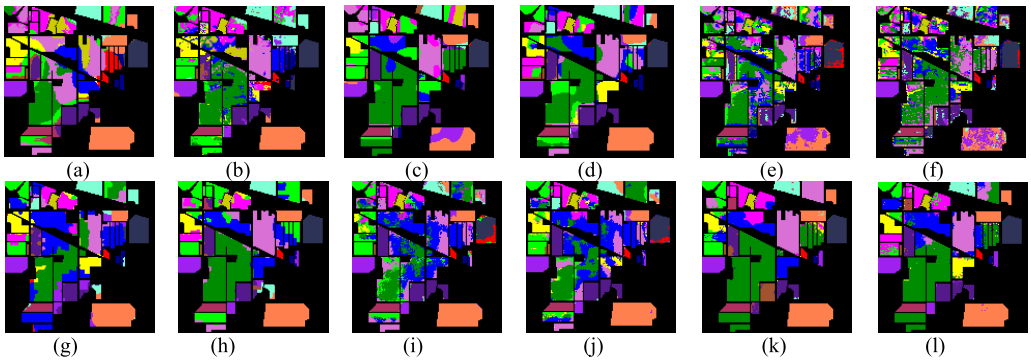


Fig. 8. Classification maps of different methods on the Indian Pines dataset. (a) GCN. (b) DCGCN. (c) S^2GCN. (d) DCS^{2}GCN. (e) FuNet-C. (f) DCFuNet-C. (g) MDGCN. (h) DCMDGCN. (i) CEGCN. (j) DCCEGCN. (k) GCGCN. (l) DCGCGCN.

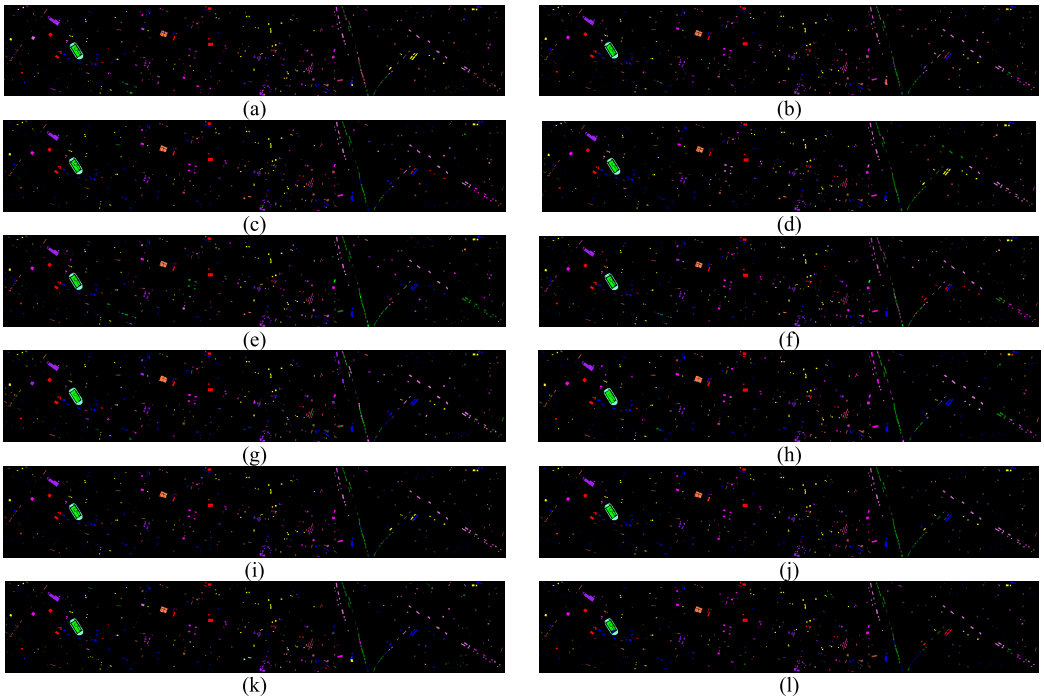


Fig. 9. Classification maps of different methods on the Houston dataset. (a) GCN. (b) DCGCN. (c) S^2GCN. (d) DCS^{2}GCN. (e) FuNet-C. (f) DCFuNet-C. (g) MDGCN. (h) DCMDGCN. (i) CEGCN. (j) DCCEGCN. (k) GCGCN. (l) DCGCGCN.

which can be obtained empirically and set to the constant one. Only the parameter η needs to be adjusted in the DCGCN model to obtain better HSI classification results. Moreover, the parameter η plays a vital role in low-pass filtering. Inspired by previous studies [53], [54], this parameter is tuned from 1 to 6 with an interval of one to observe the change of classification performance on three datasets, as shown in Fig. 12. It can be observed that the classification performance of the DCGCN method first increases to a peak value and then decrease over the three datasets. This result suggests that

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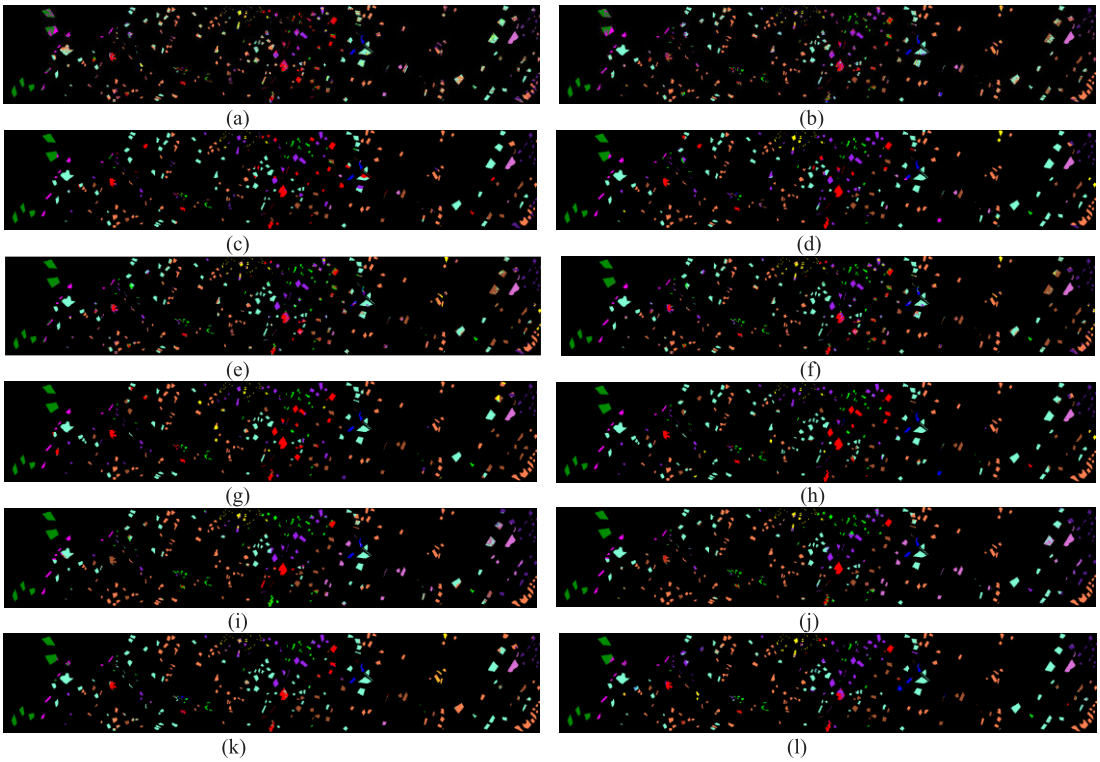


Fig. 10. Classification maps of different methods on the HyRANK dataset. (a) GCN. (b) DCGCN. (c) S2GCN. (d) DCS2GCN. (e) FuNet-C. (f) DCFuNet-C. (g) MDGCN. (h) DCMDGCN. (i) CEGCN. (j) DCCEGCN. (k) GCGCN. (l) DCGCGCN.

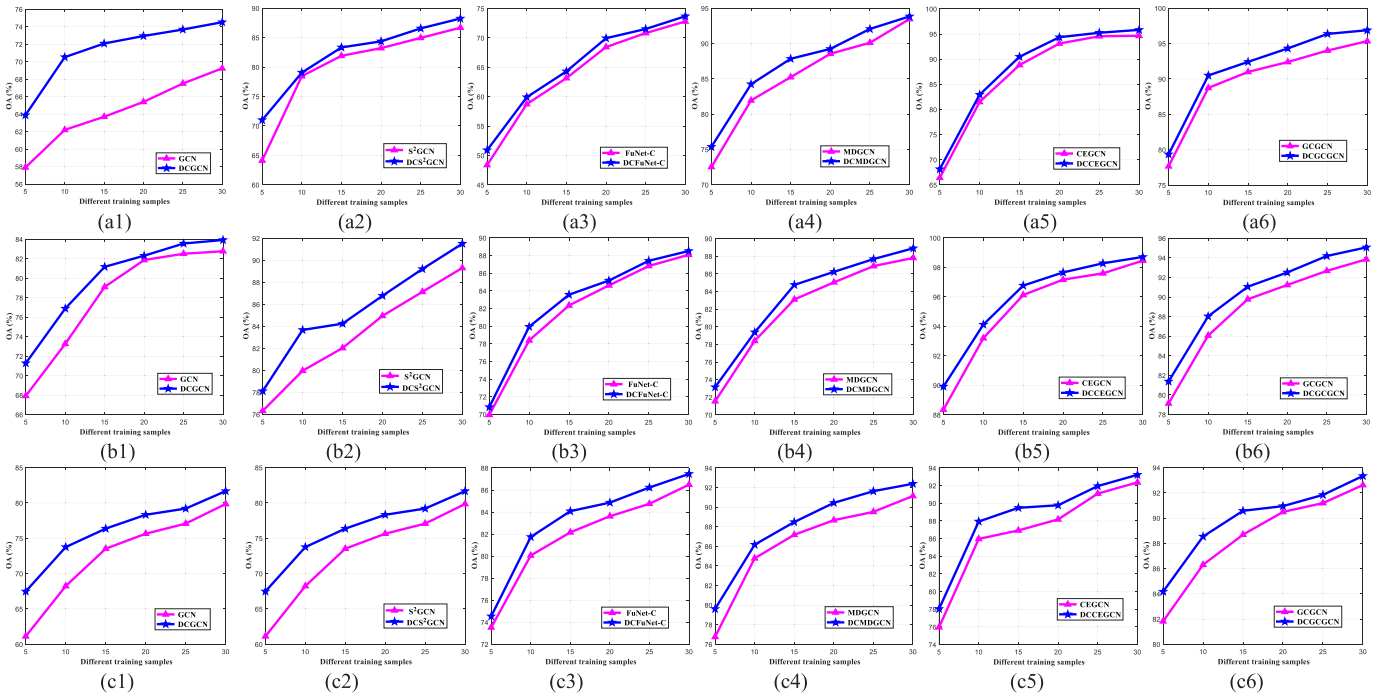


Fig. 11. OA of different methods on three datasets with different training samples. (a1)–(a6) Denote Indian Pines. (b1)–(b6) Denote Houston. (c1)–(c6) Denote HyRANK.

the selection of the parameter η is within a reasonable range. When the parameter η is large, the DCGCN method cannot learn more information from surrounding neighbors because the core node is dominant in feature propagation. When parameter η is small, the DCGCN method can learn more from surrounding neighbors because the core node can enrich itself by performing the feature propagation of diverse neighbors. Thus, the parameter η is set as two to achieve low-pass filtering.

G. Analysis of Clustering Performance on Graph

In this section, in order to evaluate the quality of graph structure from another perspective, the spectral clustering performance with graph structure is used as the criteria to evaluate

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TABLE VI
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THREE DATASETS

Data	Indian Pines			Houston 2013			HyRANK		
Class	GCN	DGCN	DCGCN	GCN	DGCN	DCGCN	GCN	DGCN	DCGCN
1	100(0)	98.37(15.98)	94.63(3.18)	100(0)	91.78(31.28)	85.61(4.46)	79.41(5.72)	39.87(0.50)	70.30(8.62)
2	31.23(16.58)	49.17(9.02)	**42.22**(12.14)	31.23(16.58)	77.45(13.10)	**90.54**(8.12)	62.98(18.60)	83.31(18.48)	**80.91**(1.42)
3	18.55(18.26)	53.58(12.08)	**52.19**(10.13)	18.55(18.26)	96.27(1.69)	**97.57**(1.28)	39.68(19.43)	62.2(24.38)	**41.96**(34.49)
4	97.49(6.16)	71.41(10.77)	79.48(7.82)	97.49(6.16)	89.70(3.56)	87.33(2.28)	84.14(10.02)	83.45(14.14)	**91.38**(4.38)
5	67.78(15.58)	68.97(13.97)	**70.71**(10.51)	67.78(15.58)	87.53(6.54)	**93.80**(5.29)	36.09(8.01)	54.43(2.96)	**43.93**(1.48)
6	61.77(6.91)	70.02(12.67)	**76.64**(19.32)	61.77(6.91)	85.75(6.51)	84.43(5.30)	96.63(4.46)	98.26(0)	90.59(1.39)
7	99.28(1.78)	100(0)	96.52(3.64)	99.28(1.78)	52.64(16.06)	**67.72**(12.95)	76.97(7.60)	71.20(4.05)	70.93(2.28)
8	100(0)	97.68(1.26)	99.92(0.19)	100(0)	50.72(10.62)	43.44(11.89)	59.62(19.19)	66.18(5.21)	**73.65**(8.29)
9	100(0)	**100(0)**	**100**(0)	100(0)	49.32(11.34)	**66.22**(10.81)	44.77(19.43)	51.87(1.01)	41.23(8.41)
10	53.09(19.95)	66.80(17.15)	**72.58**(7.08)	53.09(19.95)	67.15(12.83)	**60.79**(12.18)	89.18(1.57)	89.81(1.08)	87.47(4.36)
11	52.35(20.72)	43.81(13.30)	**59.97**(5.02)	52.35(20.72)	56.99(6.91)	**51.00**(15.85)	70.94(10.10)	75.00(0.79)	**72.13**(0.62)
12	42.43(8.85)	49.94(17.44)	**51.77**(5.76)	42.43(8.85)	48.93(7.61)	**41.63**(9.85)	95.25(3.75)	75.22(3.42)	**96.06**(4.86)
13	97.33(5.81)	99.67(16.96)	**98.80**(1.15)	97.33(5.81)	24.87(10.14)	**29.24**(4.87)			
14	88.94(6.49)	79.13(4.84)	64.76(13.81)	88.94(6.49)	96.08(11.15)	93.38(4.42)			
15	76.90(0.17)	71.48(6.70)	69.45(15.36)	76.90(0.17)	97.04(5.68)	97.99(1.51)			
16	99.05(0.86)	92.05(3.34)	95.45(4.17)						
OA	57.90(8.60)	61.44(5.53)	**63.88**(4.96)	67.93(2.20)	69.77(3.76)	**71.28**(0.91)	64.44(2.39)	65.71(2.02)	**67.48**(2.98)
AA	74.14(4.13)	75.76(3.22)	**76.57**(2.98)	70.19(2.04)	71.48(3.01)	**72.71**(0.95)	69.64(0.63)	70.9(1.63)	**71.71**(1.92)
KC	53.78(8.85)	57.08(5.76)	**59.57**(5.47)	65.40(2.37)	67.37(4.04)	**69.00**(0.96)	57.69(2.34)	58.83(2.28)	**60.96**(3.22)

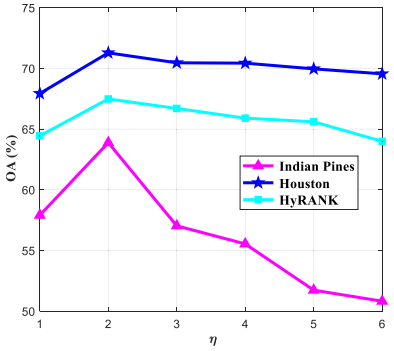


Fig. 12. Classification performance of the DCGCN method with different parameter values on three HSI datasets.

the quality of graph structure on three datasets including Indian Pines, Houston 2013, and HyRANK. Specifically, the spectral clustering aims to find a partition of the graph such that the edges between different groups have very low weights (which means that points in different clusters are dissimilar from each other) and the edges within a group have high weights (which means that points within the same cluster are similar to each other) [55]. Specific clustering results on three HSI datasets are shown in Table VII. In the meanwhile, the ac and NMI as evaluation indicators are used to evaluate the clustering performance. The closer they are to 100%, the better their clustering performance. From Table VII, for local graphs from GCN, S^2GCN, FuNet-C methods, their clustering results with the graph construction is less than that of the graph from DCGCN, DCS^{2}GCN, DCFuNet-C methods, which demonstrates that the DC restrictions can alleviate the influence of the noise and spectral variability for the local graph construction. For graphs with rich contextual information from MDGCN, and GCGCN methods, their graph structures are generally influenced by the spectral similarity of interclass pixels in pursing the contextual information. However, the DCMDGCN and DCGCGCN methods can alleviate the influence of interclass similarity by imposing the DC restrictions on the initial graph. Overall, the clustering results on three datasets demonstrate that the spectral clustering performance of the improved graph structure is better than that of the initial graph structure, which proves the effectiveness of the proposed method.

H. Broad Comparison

For further testing the effectiveness of the DCGCN method, the latest popular GCN-based methods including MLGN [41], MSSGU [42], SAGN [45], F^{2}HNN [47], and AMGCFN [43], are selected to prove its effectiveness of DC restrictions. These methods are including superpixel-based graph neural networks and hypergraph-level neural networks, which are also influenced by the interclass similarity in the construction of the graph structure due to complex spectral characteristics. The corresponding experiments are conducted on six remote sensing datasets including hyperspectral remote sensing images (Indian Pines, Houston 2013, and HyRNK) and high-resolution remote sensing images (UC Merced, AID, and NWPU-RESISC45) to further demonstrate the effectiveness of the diversity and connectivity restrictions.

Experimental results are shown in Table VIII, which shows the classification results of different GCN-based methods and the improved version via our proposed DC restrictions. All the experiments are repeated five times to obtain AA. The number of the training samples is set five that is consistent with the above settings. From Table VIII, the F^{2}HNN, MLGN, MSSGU, and AMGCFN methods achieve good classification results, but the improved methods including DCF^{2}HNN, DCMLGN, DCMSSGU, and DCAMGCFN methods achieves better classification results on six remote sensing datasets compared to original classification methods. The main reason for this phenomenon is that the DC restrictions can remove the weak interclass connections and strong interclass connections of hypergraph-level and superpixel-level graph structures. And the experiments further demonstrate the effectiveness and adaptivity of the proposed plug-and-play model in the existing GCN-based methods. Among them, the SAGN method is

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TABLE VII
CLUSTERING RESULTS (%) OF DIFFERENT METHODS ON THREE DATASETS

HyRANK										
Methods	GCN	DCGCN	S^2GCN	DCS^{2}GCN	FuNet-C	DCFuNet-C	MDGCN	DCMDGCN	GCGCN	DCGCGCN
AC	32.63(6.74)	**34.68**(4.05)	37.46(3.27)	**38.33**(3.07)	35.33(4.51)	**36.94**(3.79)	53.85(3.18)	**56.43**(4.68)	62.97(2.59)	**64.86**(1.47)
NMI	32.10(5.87)	**35.89**(4.39)	31.72(2.98)	**33.99**(2.84)	39.46(4.26)	**41.29**(3.55)	61.22(3.52)	**63.53**(4.31)	67.84(2.86)	**70.03**(1.59)
Houston 2013										
	GCN	DCGCN	S^2GCN	DCS^{2}GCN	FuNet-C	DCFuNet-C	MDGCN	DCMDGCN	GCGCN	DCGCGCN
AC	31.21(5.28)	**32.69**(4.98)	36.32(2.47)	**37.41**(1.98)	33.78(1.87)	**34.80**(2.31)	39.74(1.46)	**42.57**(2.36)	52.20(1.74)	**54.17**(0.82)
NMI	36.55(4.96)	**38.05**(5.07)	45.42(2.87)	**46.31**(2.15)	42.28(1.42)	**43.21**(2.07)	37.71(1.87)	**40.72**(2.48)	50.88(1.65)	**52.53**(0.91)
Indian Pines										
	GCN	DCGCN	S^2GCN	DCS^{2}GCN	FuNet-C	DCFuNet-C	MDGCN	DCMDGCN	GCGCN	DCGCGCN
AC	24.27(4.39)	**26.53**(4.67)	34.46(3.85)	**40.78**(3.54)	30.22(4.23)	**34.45**(4.38)	43.14(3.22)	**46.32**(2.67)	48.66(1.86)	**49.89**(1.16)
NMI	22.88(4.31)	**24.67**(4.58)	30.06(3.69)	**39.23**(3.47)	31.32(4.41)	**32.84**(4.62)	42.63(3.19)	**43.57**(2.58)	43.94(1.74)	**46.78**(1.08)

*AC denotes the clustering accuracy, and NMI denotes normalized mutual information. The closer they are to 100 %, the better their clustering performance.

TABLE VIII
CLASSIFICATION RESULTS (%) OF DIFFERENT GCN-BASED METHODS ON DIFFERENT DDATASETS

Indian Pines									UC Merced	
Methods	F^2HNN	DC F^2HNN	MLGN	DCMLGN	MSSGU	DCMSSGU	AMGCFN	DC AMGCFN	SAGN	DCSAGN
OA	67.62(2.83)	**68.09**(2.37)	75.67(3.47)	**76.76**(4.96)	76.49(2.06)	**77.15**(3.90)	72.86(4.29)	**76.28**(2.36)	82.54(0.21)	**83.41**(0.24)
AA	82.23(1.25)	**84.13**(1.31)	84.27(2.35)	**85.38**(2.69)	84.34(1.92)	**85.12**(2.76)	82.74(2.56)	**84.75**(2.06)	82.48(0.21)	**83.30**(0.24)
KC	63.97(2.17)	**65.54**(1.89)	72.77(3.70)	**73.95**(5.44)	73.22(2.92)	**74.02**(3.22)	69.48(4.75)	**73.29**(2.55)	81.64(0.19)	**82.37**(0.20)
Houston 2013									AID	
Methods	F^2HNN	DCF^{2}HNN	MLGN	DCMLGN	MSSGU	DCMSSGU	AMGCFN	DC AMGCFN	SAGN	DCSAGN
OA	64.21(2.55)	**66.57**(1.27)	79.50(2.66)	**81.39**(3.24)	75.46(1.52)	**76.74**(2.13)	75.51(3.93)	**76.54**(2.77)	74.51(0.25)	**75.35**(0.23)
AA	69.30(1.77)	**71.30**(1.16)	81.99(1.30)	**83.20**(2.53)	78.67(1.47)	**79.11**(1.55)	78.49(3.32)	**80.04**(2.59)	74.42(0.24)	**75.36**(0.21)
KC	62.05(1.56)	**64.02**(1.37)	77.85(2.84)	**79.88**(3.43)	73.72(1.45)	**74.96**(1.91)	73.64(4.23)	**74.69**(2.99)	73.62(0.24)	**74.49**(0.22)
HyRANK									NWPU-RESISC45	
Methods	F^2HNN	DCF^{2}HNN	MLGN	DCMLGN	MSSGU	DCMSSGU	AMGCFN	DC AMGCFN	SAGN	DCSAGN
OA	70.92(2.17)	**72.31**(1.98)	77.12(4.51)	**78.28**(4.96)	72.01(2.19)	**73.44**(2.61)	67.09(7.52)	**68.22**(3.51)	64.39(0.18)	**65.83**(0.17)
AA	77.31(0.77)	**78.76**(0.71)	81.63(2.80)	**82.79**(2.93)	76.75(1.67)	**76.90**(2.40)	77.63(4.26)	**78.17**(2.93)	64.32(0.16)	**65.80**(0.14)
KC	66.34(2.48)	**68.97**(2.37)	72.32(5.01)	**73.63**(5.71)	65.74(3.37)	**67.7**(3.67)	60.59(8.51)	**62.02**(3.89)	63.58(0.13)	**65.04**(0.12)

mainly suitable for high-resolution remote sensing image classification. In order to prove the effectiveness of the DC restrictions, related experiments are conducted on UC Merced, AID, and NWPU-RESISC45 datasets to demonstrate that the classification performance of the DCSAGN method is better than that of the SAGN method.

VI. CONCLUSION

In this article, for fully improving the quality of the given graph structure in GCN-based methods, an effective post-processing framework DCGCN, which is independent of the network architecture, is proposed to improve the DC of the given graph structure. The quality of graph structure can be well preserved by neighbor filtering through the DC operations. The diverse restriction is to remove the weak interclass connections, and the key connectivity restriction is to remove the strong interclass connections finely based on the diverse neighbors. Finally, the effectiveness of the proposed DCGCN method is tested on three real hyperspectral datasets including Indian Pines, Houston 2013, HyRANK, to demonstrate the effectiveness of the DCGCN method and to improve the graph quality of GCN-based methods for HSI classification.

In the future, the DCGCN method can still be further improved. Since the basic idea of the diversity-connected graph is to remove the interclass connections of pixels and improve the quality of the graph structure. The working mechanism of DCGCN is not changed in the massage passing of neighboring information. In other words, the DCGCN is still facing the over-smoothing problem with the increasingly deeper graph convolution layers, which will be addressed in future work.

APPENDIX
PROOF FOR THEOREM 1

For $D_{\mathrm{KL}}(S\|C)$, since the PDFs of S and C are unknown, we use a nonparametric way, that is histogram, to estimate the PDFs of S and C. Specifically, we uniformly divide the graph feature space $\Omega = [0,1]^d$ into r^d bins $\{H_1, H_2, \ldots, H_{r^d}\}$, whose length is $(1/r^d)$ and dimension is d. For simplifying the use of notations, $|H_i|_C$ and $|H_i|_S$ denote the number of samples that are in bin H_i. Thus, we have

$$D_{\mathrm{KL}}(S\|C) \approx \sum_{i=1}^{r^d} \frac{|H_i|_S}{2|\varepsilon|} \cdot \log \frac{|H_i|_S/2|\varepsilon|}{|H_i|_C/2|\varepsilon|}$$

$$= \frac{1}{2|\varepsilon|} \cdot \sum_{i=1}^{r^d} |H_i|_S \cdot \log \frac{|H_i|_S}{|H_i|_C} = \frac{1}{2|\varepsilon|} \cdot \left(\begin{array}{l} \sum_{i=1}^{r^d} |H_i|_S \cdot \log |H_i|_S \\ -\sum_{i=1}^{r^d} |H_i|_S \cdot \log |H_i|_C \end{array} \right)$$

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$$= \frac{1}{2|\varepsilon|} \cdot \begin{pmatrix} \sum_{i=1}^{r^d} |H_i|_S \cdot \log |H_i|_S \\ -\sum_{i=1}^{r^d} |H_i|_S \cdot \log(|H_i|_S + \sigma_i) \end{pmatrix}$$

where $\sigma_i = |H_i|_C - |H_i|_S$ and σ_i denotes an independent variable. Next, for the sake of subsequent proof, we need to perform a second-order Taylor approximation with the item $\sum_{i=1}^{r^d} |H_i|_S \cdot \log(|H_i|_S + \sigma_i)$ at point zero as

$$\sum_{i=1}^{r^d} |H_i|_S \cdot \log(|H_i|_S + \sigma_i) \approx \sum_{i=1}^{r^d} |H_i|_S \cdot \left(\log |H_i|_S + \frac{\sigma_i}{|H_i|_S \ln 2} - \frac{(\sigma_i)^2}{2(|H_i|_S)^2 \ln 2} \right).$$

In fact, the number of the core node $|H_i|_C$ is identical to that of the surrounding neighbors $|H_i|_S$ in each bin H_i because the core node is connected by surrounding neighbors with edge weight in graph feature space. Thus, we have $\sum_{i=1}^{r^d} |H_i|_C - |H_i|_S = \sum_{i=1}^{r^d} \sigma_i = 0$. Based on the above analysis, we can rewrite $D_{\mathrm{KL}}(S\|C)$ as

$$\begin{aligned} &D_{\mathrm{KL}}(S\|C) \\ &\approx \sum_{i=1}^{r^d} \frac{|H_i|_S}{2|\varepsilon|} \cdot \log \frac{|H_i|_S / 2|\varepsilon|}{|H_i|_C / 2|\varepsilon|} \\ &= \frac{1}{2|\varepsilon|} \cdot \begin{pmatrix} \sum_{i=1}^{r^d} |H_i|_S \cdot \log |H_i|_S \\ -\sum_{i=1}^{r^d} |H_i|_S \cdot \log(|H_i|_S + \sigma_i) \end{pmatrix} \\ &\approx \frac{1}{2|\varepsilon|} \cdot \begin{pmatrix} \sum_{i=1}^{r^d} |H_i|_S \cdot \log |H_i|_S \\ -\sum_{i=1}^{r^d} |H_i|_S \cdot \left(\log |H_i|_S + \frac{\sigma_i}{|H_i|_S \ln 2} - \frac{(\sigma_i)^2}{2(|H_i|_S)^2 \ln 2} \right) \end{pmatrix} \\ &= \frac{1}{2|\varepsilon|} \cdot \sum_{i=1}^{r^d} \left(\frac{(\sigma_i)^2}{2|H_i|_S \ln 2} - \frac{\sigma_i}{\ln 2} \right) \\ &= \frac{1}{4 \ln 2 |\varepsilon|} \cdot \sum_{i=1}^{r^d} \frac{(\sigma_i)^2}{|H_i|_S}. \end{aligned}$$

In fact, $D_{\mathrm{KL}}(S\|C)$ is closely related to the independent variable σ_i, and σ_i denotes the Chi-Square distance between $|H_i|_C$ and $|H_i|_S$. It can be observed that the effective diverse information $D_{\mathrm{KL}}(S\|C)$ will be large when σ_i is large if the $|H_i|_S$ is considered as a constant. Suppose that the number of surrounding neighbors is $|\mathbb{N}(v_i)|$, we have the expectation of $|H_i|_C$ and $|H_i|_S$ is $\mathrm{E}(|H_i|_C - |H_i|_S) = \mathrm{E}(|\mathbb{N}(v_i)| \cdot \mathbf{h}_i - \sum_{j \in \mathbb{N}(v_i)} \mathbf{h}_j) = 0$ where $\mathrm{E}(\cdot)$ denotes the expectation. For their variance, it can be described as follows:

$$\begin{aligned} &\mathrm{Var}(|H_i|_C - |H_i|_S) \\ &= \mathrm{Var}\left(|\mathbb{N}(v_i)| \cdot \mathbf{h}_i - \sum_{j \in \mathbb{N}(v_i)} \mathbf{h}_j \right) \\ &= \mathrm{E}\left(\left(|\mathbb{N}(v_i)| \cdot \mathbf{h}_i - \sum_{j \in \mathbb{N}(v_i)} \mathbf{h}_j \right)^2 \right) \\ &\quad - \mathrm{E}\left(|\mathbb{N}(v_i)| \cdot \mathbf{h}_i - \sum_{j \in \mathbb{N}(v_i)} \mathbf{h}_j \right) \\ &= \mathrm{E}\left(\left(|\mathbb{N}(v_i)| \cdot \mathbf{h}_i - \sum_{j \in \mathbb{N}(v_i)} \mathbf{h}_j \right)^2 \right) = \mathrm{E}\big((\sigma_i)^2\big) \geq 0. \end{aligned}$$

Meanwhile, the item $(|\mathbb{N}(v_i)| \cdot \mathbf{h}_i - \sum_{j \in \mathbb{N}(v_i)} \mathbf{h}_j)^2$ can be considered as an indicator to measure the difference between the core node and the surrounding neighbors. That is to say, $D_{\mathrm{KL}}(S\|C) \approx (1/(4 \ln 2 |\varepsilon|)) \cdot \sum_{i=1}^{r^d} ((\sigma_i)^2)/(|H_i|_S)$ is positively correlated with $\mathrm{E}((|\mathbb{N}(v_i)| \cdot \mathbf{h}_i - \sum_{j \in \mathbb{N}(v_i)} \mathbf{h}_j)^2)$. The effective diverse information $D_{\mathrm{KL}}(S\|C)$ is large when the difference of their PDFs between the core node and the surrounding neighbors is large. In fact, when the core node v_i and surrounding neighbor v_j are independent, we have

$$\begin{aligned} D_{\mathrm{KL}}(S\|C) &\approx \frac{1}{4 \ln 2 |\varepsilon|} \cdot \sum_{i=1}^{r^d} \frac{(\sigma_i)^2}{|H_i|_S} \\ &\sim \mathrm{E}\left(\left(|\mathbb{N}(v_i)| \cdot \mathbf{h}_i - \sum_{j \in \mathbb{N}(v_i)} \mathbf{h}_j \right)^2 \right) \\ &\sim \mathrm{E}\left(\sum_{j \in \mathbb{N}(v_i)} \big(\mathbf{h}_i - \mathbf{h}_j \big)^2 \right). \end{aligned}$$

In terms of $\sum_{j \in \mathbb{N}(v_i)} (\mathbf{h}_i - \mathbf{h}_j)^2$, it measures the distance between the core node v_i and surrounding neighbor v_j. According to [56], the samples with a smaller distance should have a large probability similarity to be neighbors, which can be defined as follows:

$$\min_{\mathbf{w}_i^{\mathrm{T}} \mathbf{1} = 1, 0 \leq \mathbf{w}_i \leq 1} \sum_{j \in \mathbb{N}(v_i)} \left(\big(\mathbf{h}_i - \mathbf{h}_j \big)^2 w_{ij} + \lambda w_{ij}^2 \right)$$

where w_{ij} denotes the similarity between the core node v_i and surrounding neighbor v_j. The similarity w_{ij} is small when the distance $\sum_{j \in \mathbb{N}(v_i)} (\mathbf{h}_i - \mathbf{h}_j)^2$ is large. That is to say, the item $\sum_{j \in \mathbb{N}(v_i)} (\mathbf{h}_i - \mathbf{h}_j)^2$ is negatively correlated with adjacency similarity. In other words, effective diverse information $D_{\mathrm{KL}}(S\|C)$ is negatively correlated with adjacency similarity.

Recall the definition of $\lambda_a = \arg\min_{l_i} ((\sum_i \|\mathbf{W}^{l_i}\|_1)/|\varepsilon|)$ where λ_a denotes the statistical minimum distance in the connected intraclass samples with identical labels. Therefore, we can obtain $D_{\mathrm{KL}}(S\|C) \approx (1/(4 \ln 2 |\varepsilon|)) \cdot \sum_{i=1}^{r^d} ((\sigma_i)^2)/(|H_i|_S)$ is negatively correlated with the statistical threshold value λ_a.

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REFERENCES

[1] J. M. Bioucas-Dias and J. M. P. Nascimento, "Hyperspectral subspace identification," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 8, pp. 2435–2445, Aug. 2008.

[2] Y. Shi, J. Li, Y. Zheng, B. Xi, and Y. Li, "Hyperspectral target detection with RoI feature transformation and multiscale spectral attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5071–5084, Jun. 2021.

[3] D. Hong, J. Hu, J. Yao, J. Chanussot, and X. X. Zhu, "Multimodal remote sensing benchmark datasets for land cover classification with a shared and specific feature learning model," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 68–80, Aug. 2021.

[4] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[5] C. McCann, K. S. Repasky, M. Morin, R. L. Lawrence, and S. Powell, "Using Landsat surface reflectance data as a reference target for multiswath hyperspectral data collected over mixed agricultural rangeland areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5002–5014, Sep. 2017.

[6] X. Kang, S. Li, L. Fang, M. Li, and J. A. Benediktsson, "Extended random Walker-based classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 144–153, Jan. 2015.

[7] X. Kang, X. Xiang, S. Li, and J. A. Benediktsson, "PCA-based edge-preserving features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7140–7151, Dec. 2017.

[8] Q. Zhu, Y. Zhong, L. Zhang, and D. Li, "Adaptive deep sparse semantic modeling framework for high spatial resolution image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6180–6195, Oct. 2018.

[9] Y. Ding, S. Pan, and Y. Chong, "Robust spatial–spectral block-diagonal structure representation with fuzzy class probability for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1747–1762, Mar. 2020.

[10] W. Zhao and S. Du, "Spectral–spatial feature extraction for hyperspectral image classification: A dimension reduction and deep learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4544–4554, Aug. 2016.

[11] A. Villa, J. A. Benediktsson, J. Chanussot, and C. Jutten, "Hyperspectral image classification with independent component discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 4865–4876, Dec. 2011.

[12] Q. Wang, X. He, and X. Li, "Locality and structure regularized low rank representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 911–923, Feb. 2019.

[13] C. Pan, X. Jia, J. Li, and X. Gao, "Adaptive edge preserving maps in Markov random fields for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8568–8583, Oct. 2021.

[14] S. Jia, S. Jiang, Z. Lin, N. Li, M. Xu, and S. Yu, "A survey: Deep learning for hyperspectral image classification with few labeled samples," *Neurocomputing*, vol. 448, pp. 179–204, Aug. 2021.

[15] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jan. 2015.

[16] N. He et al., "Feature extraction with multiscale covariance maps for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 755–769, Feb. 2019.

[17] D. Wang, B. Du, L. Zhang, and Y. Xu, "Adaptive spectral–spatial multiscale contextual feature extraction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2461–2477, Mar. 2021.

[18] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[19] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019.

[20] R. Hang, Z. Li, Q. Liu, P. Ghamisi, and S. S. Bhattacharyya, "Hyperspectral image classification with attention-aided CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2281–2293, Mar. 2021.

[21] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[22] J. Yang, Y.-Q. Zhao, and J. C. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.

[23] Q. Liu, L. Xiao, J. Yang, and J. C. Chan, "Content-guided convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6124–6137, Sep. 2020.

[24] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2020.

[25] A. Qin, Z. Shang, J. Tian, Y. Wang, T. Zhang, and Y. Y. Tang, "Spectral–spatial graph convolutional networks for semisupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 241–245, Feb. 2019.

[26] Y. Ding, Y. Guo, Y. Chong, S. Pan, and J. Feng, "Global consistent graph convolutional network for hyperspectral image classification," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–16, 2021.

[27] L. Mou, X. Lu, X. Li, and X. X. Zhu, "Nonlocal graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8246–8257, Dec. 2020.

[28] S. Wan, C. Gong, P. Zhong, S. Pan, G. Li, and J. Yang, "Hyperspectral image classification with context-aware dynamic graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 597–612, Jan. 2021.

[29] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel- and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.

[30] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[31] X. Gao, W. Hu, and Z. Guo, "Exploring structure-adaptive graph learning for robust semi-supervised classification," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2020, pp. 1–6.

[32] W. He et al., "Non-local meets global: An integrated paradigm for hyperspectral image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 2089–2107, Apr. 2022.

[33] D. Luo et al., "Learning to drop: Robust graph neural network via topological denoising," in *Proc. 14th ACM Int. Conf. Web Search Data Mining*, Mar. 2021, pp. 779–787.

[34] U. von Luxburg, "A tutorial on spectral clustering," *Statist. Comput.*, vol. 17, no. 4, pp. 395–416, Dec. 2007.

[35] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.

[36] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Proc. NIPS*, 2016, pp. 3844–3852.

[37] J. Bruna, W. Zaremba, A. Szlam, and Y. LeCun, "Spectral networks and locally connected networks on graphs," in *Proc. Int. Conf. Learn. Represent.*, 2014, pp. 1–14.

[38] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Represent.*, 2017, pp. 1–14.

[39] P. Velichovic, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 1–12.

[40] B. Jiang, Z. Zhang, D. Lin, J. Tang, and B. Luo, "Semi-supervised learning with graph learning-convolutional networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11305–11312.

[41] S. Wan et al., "Multi-level graph learning network for hyperspectral image classification," *Pattern Recognit.*, vol. 129, Sep. 2022, Art. no. 108705.

[42] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "Multilevel superpixel structured graph U-Nets for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516115.

[43] H. Zhou, F. Luo, H. Zhuang, Z. Weng, X. Gong, and Z. Lin, "Attention multihop graph and multiscale convolutional fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508614.

[44] Y. Ding et al., "AF2GNN: Graph convolution with adaptive filters and aggregator fusion for hyperspectral image classification," *Inf. Sci.*, vol. 602, pp. 201–219, Jul. 2022.

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[45] Y. Yang, X. Tang, Y.-M. Cheung, X. Zhang, and L. Jiao, "SAGN: Semantic-aware graph network for remote sensing scene classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1011–1025, 2023.
[46] Y. Duan, F. Luo, M. Fu, Y. Niu, and X. Gong, "Classification via structure-preserved hypergraph convolution network for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507113.
[47] Z. Ma, Z. Jiang, and H. Zhang, "Hyperspectral image classification using feature fusion hypergraph convolution neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517314.
[48] Q. Li, Z. Han, and X.-M. Wu, "Deeper insights into graph convolutional networks for semi-supervised learning," in *Proc. 32nd AAAI Conf. Artif. Intell.*, 2018, pp. 1–8.
[49] Q. Wang, M. Chen, and X. Li, "Quantifying and detecting collective motion by manifold learning," in *Proc. 31st AAAI Conf. Artif. Intell.*, 2017, pp. 1–7.
[50] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How poweful are graph neural networks," in *Proc. Int. Conf. Learn. Represent.*, 2019, pp. 1–17.
[51] S. Kullback and R. A. Leibler, "On information and sufficiency," *Ann. Math. Statist.*, vol. 22, no. 1, pp. 79–86, 1951.
[52] D. I. Shuman, S. K. Narang, P. Frossard, A. Ortega, and P. Vandergheynst, "The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains," *IEEE Signal Process. Mag.*, vol. 30, no. 3, pp. 83–98, May 2013.
[53] X. Zhang, H. Liu, Q. Li, and X.-M. Wu, "Attributed graph clustering via adaptive graph convolution," in *Proc. 29th Int. Joint Conf. Artif. Intell.*, Aug. 2019, pp. 1–7.
[54] Q. Li, X.-M. Wu, H. Liu, X. Zhang, and Z. Guan, "Label efficient semi-supervised learning via graph filtering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9574–9583.
[55] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, early access, Nov. 7, 2022, doi: 10.1109/TCYB.2022.3219855.
[56] F. Nie, X. Wang, and H. Huang, "Clustering and projected clustering with adaptive neighbors," in *Proc. 20th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2014, pp. 977–986.



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